AEX 201 - Fundamentals of Extension Education

Lecture-wise Theory Study Material

B.Sc. (Hons) Agriculture - Third Semester - Tamil Nadu Agricultural University (TNAU) - 2025 Syllabus

This handbook covers all seventeen lectures of the AEX 201 theory course in the exact order of the syllabus. Each chapter expands every sub-topic with headed sections, plain-English definitions, comparison tables, worked examples, simple text diagrams and an "Exam Focus" section of key points and model questions. Examples are drawn mainly from Indian rural conditions and Tamil Nadu, so that the theory is easy to recall and to apply in the examination.

How to use this material

- Read the chapter once for understanding, then read only the tables, diagrams and Exam Focus before the examination.
- Every definition is written so that it can be reproduced word for word in a two-mark or five-mark answer.
- Comparison tables are the fastest revision tool. Most long answers in extension education are built by expanding one or two table rows into paragraphs.
- Worked examples show how numerical questions on diffusion rate, adoption percentage and demonstration economics are solved step by step.
- Tamil Nadu and India-specific examples (KVK, Uzhavan app, e-Velanmai, m-Velanmai, TNAU schemes) are used throughout because examiners reward local, concrete illustrations.

Note on scheme years and figures: where a programme's exact year or outlay is not stated here, it has been described in general terms on purpose. In the examination, always write the correct full form of an acronym first, then the year if you are certain of it. A correct general description is safer than a wrong date.

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3 Extension Systems in India - pre-independence and post-independence efforts, Reorganised Extension System (T&V).

4 Extension and Agriculture Development Programmes of ICAR / Government of India; social justice, poverty alleviation and women development programmes.

5 New Trends in Agriculture Extension - privatisation, cyber extension / e-extension, e-Velanmai, m-Velanmai, multimedia.

6 Market-led Extension and Farmer-led Extension; Expert Systems.

7 Rural Development - concept and programmes (DWCRA, MGNREGA, RSVY, IAY, SGRY, ITDP, PMGSY).

8 Mid Semester Examination.

9 Community Development - meaning, definition, concept, principles and philosophy.

10 Rural Leadership - concept, definition, types and methods of identification.

11 Extension Administration; Monitoring and Evaluation.

12 Transfer of Technology - concepts and models; Capacity Building and Training.

13 Extension Teaching Methods - meaning, classification; Individual Contact Methods.

14 Group Contact Methods.

15 Mass Contact Methods; ICT applications in Transfer of Technology.

16 Principles and Functions of Communication; Agricultural Journalism (print media).

17 Diffusion and Adoption of Innovation.

Glossary and References

A glossary of key terms and a list of standard reference texts appear at the end of the handbook.

Chapter 1. Education and Extension Education

1.1 Education: meaning

Education is the process of bringing about desirable changes in the knowledge, skills and attitudes of a person so that he or she can live a fuller and more useful life. In its widest sense education is a lifelong process that begins at birth and continues until death. It is not confined to schools, colleges or certificates; it includes everything a person learns from the family, the community, the farm, the market and the media.

In simple words, education means the deliberate and systematic development of the whole personality of an individual - head (knowledge), hand (skill) and heart (attitude). The word itself comes from the Latin words *educare* (to bring up, to nourish) and *educere* (to lead out). Both ideas are important: education nourishes the learner and also draws out the abilities that are already within the learner.

The main characteristics of education are:

- It is a lifelong and continuous process, not a one-time event.
- It brings about change - in knowledge (knowing), skill (doing) and attitude (feeling).
- It is deliberate and purposive; it has aims and objectives.
- It is both formal and informal; it happens in classrooms and also in fields and homes.
- It is a means of social change and social control.
- It develops the individual and, through the individual, the society.

1.2 Education: definitions

Different thinkers have defined education in different ways. The definitions below are commonly quoted in extension education.

- **Mahatma Gandhi:** "Education is the all-round drawing out of the best in the child and man - body, mind and spirit."
- **Swami Vivekananda:** Education is the manifestation of the perfection already in man.
- **John Dewey:** Education is the process of living through a continuous reconstruction of experiences. It is life itself, not merely a preparation for life.
- **Pestalozzi:** Education is the natural, harmonious and progressive development of man's innate powers.
- **T. Raymont:** Education is the deliberate and systematic influence exerted by the mature person upon the immature, through instruction, discipline and harmonious development of physical, intellectual, aesthetic, social and spiritual powers of the human being.
- **Redden:** Education is the deliberate and systematic influence of the mature on the immature, directed towards ends of social efficiency and individual development.

From these definitions it is clear that education is a deliberate, systematic and continuous process aimed at the all-round development of the individual and of society.

1.3 Education: types

Education is broadly classified into three types - formal, informal and non-formal. This classification is a favourite examination question and is best remembered through a comparison table.

Type	Where it happens	Who teaches	Nature	Certification	Example
Formal	Schools, colleges, universities	Trained, paid teachers	Rigid, fixed syllabus, fixed time, hierarchical, uniform	Yes - degrees and diplomas	B.Sc. Agriculture at TNAU
Informal	Family, neighbourhood, society, work place	Parents, elders, friends, peers	Unplanned, incidental, lifelong, no fixed syllabus	No certificate	Learning mother tongue, farm customs, cooking
Non-formal	Outside the formal system - camps, farmer field schools, KVK trainings, extension programmes	Extension workers, resource persons, trainers	Organised and planned, but flexible; voluntary; learner-centred; no rigid rules	Usually a certificate of participation	KVK training on drip irrigation, e-Velanmai advisory

Formal education is highly institutionalised. It has a fixed syllabus, fixed timetable, prescribed textbooks, examinations and degrees. It is largely book-centred and often removes the learner from the real-life situation.

Informal education is the natural, unplanned learning that takes place throughout life through experience, observation and imitation. It has no teacher, no syllabus and no certificate. It is the oldest form of education.

Non-formal education is the middle path. It is organised and planned like formal education but flexible and life-centred like informal education. **Extension education is essentially non-formal education** because it is carried to farmers in their own fields and homes, at times convenient to them, in a flexible manner.

> Remember the link: Extension education = non-formal education for out-of-school rural people. This single sentence often carries two or three marks.

1.4 Extension education: meaning

The word "extension" means to extend, that is, to stretch out or to reach out. In education, extension means carrying the message of education beyond the walls of the classroom and the campus, to the people who are not able to come to the institution. Extension education therefore means the education of the people, for the people, by the people, and it takes education to the doorsteps of the learner.

Extension education is a type of out-of-school education for rural people, both adults and youth, that uses the principles of informal education to help them improve their farm, home and community life through their own efforts. It is a two-way process: it takes scientific knowledge to the farmer and brings the farmer's problems back to the scientist.

1.5 Extension education: definitions

- **Dr. J. P. Leagans:** Extension education is an applied science consisting of content derived from research, accumulated field experiences and relevant principles drawn from the behavioural sciences, synthesised with useful technology into a body of philosophy, principles, content and methods focused on the problems of out-of-school education for adults and youth.

- **H. W. van den Ban and H. S. Hawkins:** Extension is the conscious use of communication of information to help people form sound opinions and make good decisions.
- **A. W. van den Ban (simple form):** Extension is a professional communication intervention deployed by an institution to induce change in voluntary behaviour.
- **Ensminger:** Extension education is education that helps people to help themselves.
- **Kelsey and Hearne:** Extension education is an educational process of change in the behaviour of people through the use of scientific knowledge.
- **Reddy and others:** Extension education is a process of bringing about desirable changes in the behaviour of people through education, in a non-formal and voluntary manner.
- **Savile:** Extension is the central process of helping people to improve their own lives by their own efforts.

From all these definitions, the key words of extension education are: **education, non-formal, voluntary, out-of-school, rural people, behaviour change, self-help, two-way communication**.

1.6 Scope of extension education

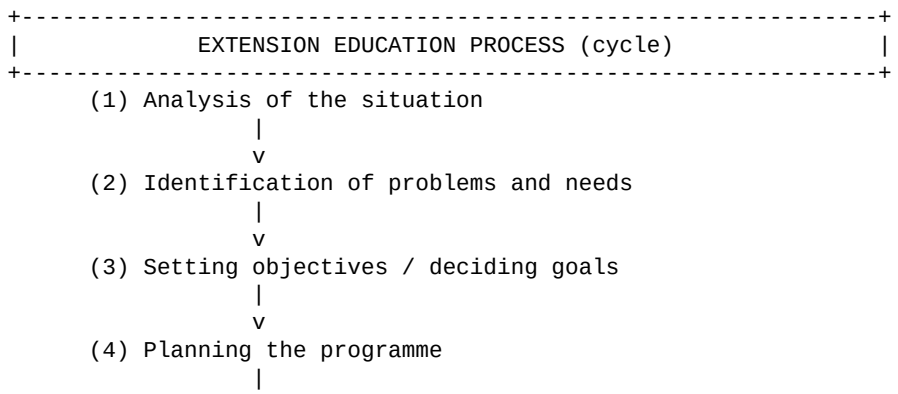
The scope of extension education means the range or the extent of its activities. Extension education today is very wide and covers every aspect of rural life. Its scope includes:

- **Agriculture and allied fields:** crop production, horticulture, livestock, fisheries, sericulture, dairying, poultry, agricultural engineering and post-harvest technology.
- **Home and family:** nutrition, health, home management, child care, family welfare.
- **Community development:** rural institutions, cooperatives, panchayats, rural leadership, rural youth work.
- **Social and cultural fields:** adult education, women empowerment, removal of social evils, social justice.
- **Economic fields:** farm management, marketing, credit, rural industries, income generation.
- **New and emerging areas:** environmental protection, organic farming, precision farming, market-led extension, cyber extension, climate-smart agriculture.

Because agriculture is the backbone of the Indian rural economy, agricultural extension occupies the largest part of the scope of extension education in India.

1.7 Extension education: process

The extension education process is a systematic series of steps by which the extension worker brings about desirable changes in the behaviour of the farmer. The process is often shown as a cycle.



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                v
(5) Execution / implementation (teaching)
                |
                v
(6) Evaluation (measure change)
                |
                v
(7) Reconsideration / modification -> back to step (1)
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The steps in brief:

1 **Analysis of the situation:** study of the people, their land, crops, resources, customs, level of technology and problems.

2 **Identification of problems and needs:** finding out what the people really need, in their own words.

3 **Objectives:** deciding what changes in knowledge, skill and attitude are wanted.

4 **Planning:** preparing a written programme of work with who, what, when, where and how.

5 **Execution:** teaching through individual, group and mass contact methods.

6 **Evaluation:** measuring whether the objectives were achieved and how much behaviour changed.

7 **Reconsideration:** using evaluation results to improve the next programme; the cycle repeats.

1.8 Objectives of extension education

Objectives are the ends towards which extension work is directed. They are usually grouped as fundamental (basic), general and specific.

Fundamental or basic objectives express the ultimate purpose of extension education. The widely quoted fundamental objectives are:

- To raise the standard of living of rural people.
- To bring about desirable changes in the behaviour of the people (knowledge, skill and attitude).
- To help people to help themselves, that is, to develop self-reliance.
- To develop the human personality and leadership of rural people.

General objectives apply to all extension work and include:

- To promote the adoption of improved agricultural technology.
- To develop rural leadership and rural institutions.
- To encourage the use of local resources for local needs.
- To improve family living, health and nutrition.
- To develop a spirit of cooperation and community feeling.

Specific objectives are the particular objectives of a single programme. For example, "to train 100 farmers of a village in the correct use of the Uzhavan mobile app during one month" is a specific objective - it is measurable and time-bound.

1.9 Principles of extension education

Principles are the broad guidelines that extension workers follow to make their work effective. The important principles are listed below.

- **Principle of change:** Extension education aims at changing the knowledge, skill and attitude of the people, and the change must come from within the learner; people cannot be forced.
- **Principle of interest and need:** Programmes must be based on the real needs and interests of the people, not on what the extension worker thinks they need.
- **Principle of participation:** People must take part in planning, executing and evaluating the programme. Participation builds ownership.
- **Principle of leadership:** Work with and through local leaders; the extension worker should develop and use local leaders.
- **Principle of cultural difference:** Recognise and respect the culture, customs, beliefs and values of the people; the programme must fit the culture.
- **Principle of whole family and whole community:** Extension must deal with the whole family (men, women and youth) and the whole community, not with a single person or a single crop.
- **Principle of cooperation and teamwork:** Extension is a team effort; specialists, administrators and local leaders must work together.
- **Principle of adaptability:** Programmes and methods must be adapted to the local situation and to the changing needs of the people.
- **Principle of "grass-roots" approach:** Programmes should start from the people at the village level and move upward, not be imposed from the top.
- **Principle of voluntary and non-formal learning:** Learning must be voluntary; teaching must be non-formal, practical and life-centred.
- **Principle of evaluation:** Progress must be measured continuously so that the programme can be corrected.
- **Principle of the two-way process:** Extension is a two-way street; the farmer's problems must travel back to the research station.

1.10 Extension education compared with formal education

A clear comparison helps in the examination.

Point of comparison	Formal education	Extension education
Learner	School and college students	Out-of-school rural adults and youth
Place	Classroom, campus	Village, farm, home
Content	Fixed, prescribed syllabus	Flexible, need-based, problem-solving
Timing	Fixed timetable, fixed duration	Convenient to the learner, continuous
Teacher	Trained, paid teacher	Extension worker / village-level worker
Method	Lecture, books, examinations	Individual, group and mass methods
Motivation	Degree, job, certificate	Improved living, higher income, self-help

Point of comparison	Formal education	Extension education
Direction	Mostly one-way (teacher to student)	Two-way (technology out, problems in)
Result	Certificate / degree	Behaviour change and better living

1.11 Worked example - classifying an activity as formal, informal or non-formal

Problem. Classify each activity and give a reason: (a) a girl learning to cook from her grandmother; (b) a farmer attending a three-day KVK training on plant protection; (c) a student writing the AEX 201 semester examination; (d) a farmer learning about the Uzhavan app from a neighbour over tea.

Solution.

Activity	Type	Reason
(a) Learning to cook from grandmother	Informal	Unplanned, no syllabus, no teacher, no certificate
(b) KVK plant protection training	Non-formal	Planned and organised, but flexible, voluntary and life-centred
(c) AEX 201 semester examination	Formal	Fixed syllabus, fixed time, institutional, leads to marks/degree
(d) Learning the app from a neighbour	Informal	Incidental, unplanned, no certification

1.12 Extension in the Indian and Tamil Nadu context

- In India, the extension machinery grew from the pre-independence rural reconstruction experiments (Sriniketan, Marthandam) to the present-day **Krishi Vigyan Kendra (KVK)** system and the **Agriculture Technology Management Agency (ATMA)**.
- In Tamil Nadu, the **Department of Agriculture** runs the block-level extension machinery, while **TNAU** runs KVKs and transfers technologies through its research stations.
- Modern Tamil Nadu examples include the **Uzhavan mobile application** (single window for crop advisories, weather, market price and machinery booking) and **e-Velanmai**, an ICT-based participatory extension advisory service that connects farmers to experts.
- These examples show how the classical principles of extension - need, participation, leadership, adaptability - continue to operate in a digital age.

Exam Focus

Key points to remember

- Education = lifelong, deliberate, systematic process of desirable change in knowledge, skill and attitude.
- Three types: formal (rigid, certified), informal (natural, uncertified), non-formal (planned but flexible) - extension is non-formal.
- Extension education = out-of-school, voluntary, non-formal education for rural people; helps people to help themselves.
- Key words of extension: education, non-formal, voluntary, two-way, behaviour change, self-help.

- Fundamental objectives: raise standard of living, behaviour change, self-help, develop personality/leadership.
- Principles: change, interest and need, participation, leadership, cultural difference, whole family, cooperation, adaptability, grass-roots, voluntary learning, evaluation, two-way process.
- Extension is a two-way process: technology goes out, farmers' problems come back.

Short-answer questions with model answers

Q1. Define education and give any two definitions.

Ans. Education is the deliberate and systematic process of bringing about desirable changes in the knowledge, skill and attitude of a person for a fuller life. Gandhi: "all-round drawing out of the best in the child and man - body, mind and spirit." Dewey: "education is the continuous reconstruction of experiences."

Q2. What is extension education? Give one standard definition.

Ans. Extension education is out-of-school, non-formal education for rural people carried to their doorstep. Leagans defined it as an applied science consisting of content derived from research and field experience and principles from behavioural science, focused on the problems of out-of-school education of adults and youth.

Q3. State any four principles of extension education.

Ans. (i) Principle of need and interest; (ii) principle of participation; (iii) principle of leadership (work through local leaders); (iv) principle of cultural difference; also adaptability, whole-family approach and the two-way process.

Q4. Differentiate formal, informal and non-formal education.

Ans. Formal: institutional, fixed syllabus/time, certified (e.g. B.Sc. Agriculture). Informal: unplanned, lifelong, from family and society, uncertified (mother tongue). Non-formal: organised but flexible, voluntary, learner-centred, uncertified or lightly certified (KVK training).

Long-answer questions with model answer outlines

Q1. Explain the meaning, scope, process, objectives and principles of extension education.

Outline. Define education and extension education with two definitions each; state that extension is non-formal education. Scope: agriculture and allied, home and family, community development, social and economic fields, new areas. Process: seven steps - situation analysis, need identification, objectives, planning, execution, evaluation, reconsideration (draw the cycle). Objectives: fundamental, general, specific with one example each. Principles: list and briefly explain ten to twelve principles. Conclude with the two-way nature of extension and one Tamil Nadu example (KVK / Uzhavan).

Q2. "Extension education is non-formal education." Justify with suitable examples and compare it with formal education.

Outline. Explain the three types of education with the comparison table. Show that extension has an organised plan (like formal) but is flexible, voluntary and life-centred (like informal); it is carried to the village and farm; it has no degree. Give examples: KVK training and e-Velanmai as non-formal; B.Sc. Agriculture as formal; learning mother tongue as informal. Conclude that extension is thus a distinct non-formal system.

Chapter 2. Extension Programme Planning

2.1 Meaning of extension programme planning

An extension programme is a written statement of the situation, the objectives, the activities to be carried out and the methods to be used to bring about desirable changes among the people. Programme planning is the process of deciding in advance what is to be done, who is to do it, when, where and how it is to be done, and how the results will be measured.

Programme planning is not a one-time act; it is a continuous and democratic process in which the people themselves take part along with the extension worker. It converts vague hopes into a definite plan of action.

2.2 Definitions of programme planning

- **Kelsey and Hearne:** Programme planning is a process of determining the objectives, establishing a course of action to be followed, deciding the methods and activities to be used, and evaluating the results.
- **Boone:** Extension programme planning is a process through which an organisation, in cooperation with the people, determines and uses educational means to help people improve their own situation.
- **Boyle:** A programme is a plan of work with a definite objective, a course of action and a method of evaluation.
- **Reddy:** Programme planning is the process of analysing the situation, determining objectives, deciding the plan of work and evaluating the progress.

From these definitions, the essential elements of programme planning are: **situation, objectives, plan of work, methods, and evaluation**.

2.3 Objectives and importance of programme planning

The importance of programme planning is best remembered as a list of benefits.

- It gives a clear direction and purpose to extension work; work is not done blindly.
- It ensures that the programme is based on the real needs of the people.
- It makes the best use of limited resources - money, time, staff and materials.
- It avoids duplication of effort and overlapping of work among departments.
- It provides a basis for cooperation and coordination among agencies.
- It provides a yardstick (benchmark) for evaluation; without a plan there is nothing to measure against.
- It secures the participation and interest of the people, which ensures their support.
- It helps in the continuity of work from year to year.

2.4 Principles of programme planning

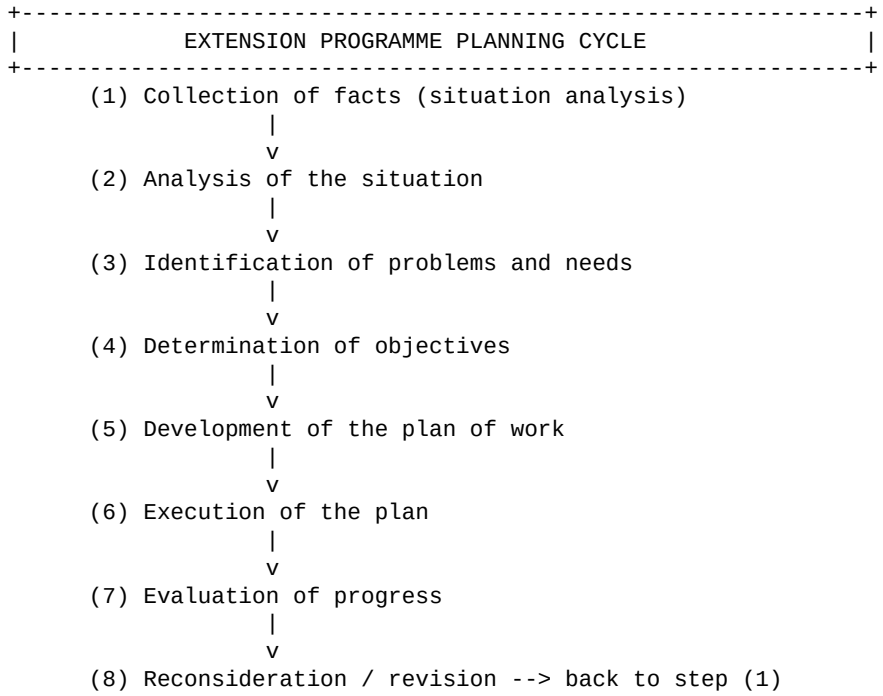
The principles of programme planning are the guidelines that make a plan sound and workable.

- **Principle of need and interest:** The programme must be based on the real needs and interests of the people.

- **Principle of the whole family and whole community:** The plan should cover the whole family and the whole village, not one individual or one enterprise.
- **Principle of participation:** People should take part in planning; a programme made by the people is a programme for the people.
- **Principle of local leadership:** Local leaders must be involved in planning and in carrying out the plan.
- **Principle of flexibility:** The plan must be flexible enough to be changed as conditions change.
- **Principle of cooperation and coordination:** All agencies - agriculture, animal husbandry, cooperation, banks, panchayat - should plan together.
- **Principle of definite and clear objectives:** Objectives must be clear, specific, achievable and measurable.
- **Principle of evaluation:** Provision for continuous evaluation must be built into the plan.
- **Principle of realism:** The plan must be within the resources, ability and culture of the people.
- **Principle of balance:** There must be a balance between different activities (crop, livestock, home, community).

2.5 Steps in programme planning (programme development)

Programme planning follows a logical sequence of steps. The sequence is often shown as a cycle.



The steps are explained below.

1. **Collection of facts:** Facts about the village are collected - population, land, crops, livestock, income, health, education, institutions, customs, resources and communication facilities. Facts may be collected by personal visits, surveys, records and group discussions.
2. **Analysis of the situation:** The facts collected are analysed to understand the real situation. A situation analysis describes what the situation is now and what it could be.

3 **Identification of problems and needs:** From the analysis, the problems and needs of the people are listed and arranged in order of importance. The people themselves should state their problems.

4 **Determination of objectives:** Clear objectives are fixed - what changes are to be brought about, in whom, and by when. Objectives should be specific, measurable, achievable, relevant and time-bound.

5 **Development of the plan of work:** A written plan of work is prepared showing the activities, the people responsible, the time schedule, the resources needed and the methods to be used. This is the heart of programme planning.

6 **Execution of the plan:** The plan is carried out through teaching - using individual, group and mass contact methods. Local leaders and people are involved at every stage.

7 **Evaluation of progress:** Progress is measured against the objectives. What was achieved? What was not? Why?

8 **Reconsideration / revision:** The results of evaluation are used to revise the plan. The revised plan becomes the basis for the next cycle.

2.6 Contents of a plan of work

A good plan of work contains the following items. This list is often asked as a short-answer question.

- Title and the area covered.
- Statement of the situation.
- Objectives - general and specific.
- The activities and the sequence of activities.
- The people responsible for each activity (who).
- The time schedule (when).
- The place (where).
- The methods and materials to be used (how).
- The resources required and their source.
- The provision for evaluation.

2.7 Worked example - building a plan of work for a village

Problem. A village-level extension worker in a Coimbatore district village finds that paddy yield is low because farmers transplant late and use excess nitrogen. Prepare a simple plan of work for one season.

Solution.

Step	Content for this village
Situation	120 ha paddy; average yield 4.2 t/ha against potential 6.0 t/ha; late transplanting and excess N common
Problem	Late transplanting and excess nitrogen reduce yield and raise cost
Objective	To raise average paddy yield by 15% in one season through timely transplanting and balanced fertiliser use

Step	Content for this village
Activity 1	Group meeting to explain the problem and the recommended package
Activity 2	Method demonstration of balanced fertiliser application
Activity 3	Result demonstration in two fields with a progressive farmer
Activity 4	Farm and home visits to 30 late-transplanting farmers
Methods	Group meeting, method demonstration, result demonstration, farm and home visit
Time	One season (June to October)
Responsible	Village Extension Worker with the Assistant Agriculture Officer and one local leader
Evaluation	Compare yield of demonstration plots with control plots; count number of farmers who transplanted on time

2.8 Difference between a programme and a plan of work

Point	Extension programme	Plan of work
Scope	Broad; covers the whole situation and long-term aims	Specific; the activities for a definite period
Duration	Long-term, continuing	Short-term, usually one year or one season
Content	Situation, needs, objectives, general approach	Activities, time schedule, methods, responsibilities
Relation	The programme is the "what" and "why"	The plan of work is the "how", "when" and "who"
Example	To improve the living standards of the village	To hold four demonstrations on balanced fertiliser in Kharif

Exam Focus

Key points to remember

- Programme = written statement of situation, objectives, activities and methods; planning = deciding in advance what, who, when, where, how.
- Programme planning is continuous and democratic, with people's participation.
- Essential elements: situation, objectives, plan of work, methods, evaluation.
- Eight steps: fact collection, situation analysis, need identification, objectives, plan of work, execution, evaluation, reconsideration.
- Plan of work contains activities, responsibility, time, place, methods, resources and evaluation provision.
- Principles: need and interest, whole family, participation, local leadership, flexibility, cooperation, clear objectives, evaluation, realism, balance.
- Programme (broad, long-term) differs from plan of work (specific, short-term).

Short-answer questions with model answers

Q1. Define extension programme planning.

Ans. Programme planning is the process of determining objectives, establishing a course of action, deciding the methods and activities to be used, and evaluating the results, with the participation of the

people (Kelsey and Hearne).

Q2. List the eight steps in programme planning.

Ans. (1) Collection of facts; (2) analysis of the situation; (3) identification of problems and needs; (4) determination of objectives; (5) development of the plan of work; (6) execution of the plan; (7) evaluation of progress; (8) reconsideration and revision.

Q3. What are the contents of a plan of work?

Ans. Title and area, statement of situation, objectives, activities and their sequence, persons responsible, time schedule, place, methods and materials, resources, and provision for evaluation.

Q4. State any four principles of programme planning.

Ans. (i) Need and interest of the people; (ii) participation of the people; (iii) flexibility; (iv) provision for evaluation. Others: local leadership, cooperation, clear objectives, realism and balance.

Long-answer questions with model answer outlines

Q1. Explain the meaning, importance, principles and steps of extension programme planning.

Outline. Define programme and programme planning with two definitions. Importance: direction, need-based work, best use of resources, avoids duplication, basis for coordination and evaluation, people's participation, continuity. Principles: list and explain ten. Steps: the eight steps with the cycle diagram; explain each step in two or three lines. Give a village plan-of-work example.

Q2. Distinguish between an extension programme and a plan of work. Illustrate with an example.

Outline. Define both. Present the comparison table (scope, duration, content, relation). Then give an example: the programme is to improve living standards of the village; the plan of work is to hold four demonstrations on balanced fertiliser in the Kharif season. Conclude that the plan of work is the yearly working tool of the programme.

Chapter 3. Extension Systems in India

3.1 Early extension efforts in India

Organised extension work in India began long before independence, through rural reconstruction experiments led by social reformers and administrators. These experiments tried to improve the whole life of the village - agriculture, health, education, sanitation and rural industries - by working with the people. The four most important pre-independence efforts are Sriniketan, Marthandam, the Firka Development Scheme and the Gurgaon Experiment.

3.2 Pre-independence extension efforts

3.2.1 Sriniketan (West Bengal)

Sriniketan was started by **Rabindranath Tagore** in 1921 near Santiniketan in West Bengal as a centre for rural reconstruction. It aimed at the all-round development of the village through education, agriculture, health, cooperatives and rural industries. Trained workers lived with the villagers, studied their problems and helped them to solve those problems by their own efforts. Sriniketan is remembered for pioneering the idea that rural development must be based on the real needs of the people and on village self-reliance.

3.2.2 Marthandam (Tamil Nadu)

The Marthandam Rural Reconstruction Centre was started in 1921 by **Dr. Spencer Hatch**, an American missionary, in the Kanyakumari (then Travancore) area of present-day Tamil Nadu. Its motto was "**Self-help with the aid of a friendly hand**". The project worked on the "three-fold programme" of the body, mind and spirit - economic development (agriculture, animal husbandry, rural industries), social development (education, health, recreation) and spiritual development. Marthandam is important for Tamil Nadu students because it is the earliest well-known rural reconstruction experiment in the region and demonstrated the principle of self-help.

3.2.3 Firka Development Scheme (Madras Presidency)

The Firka Development Scheme was launched by the **Madras (Government) in 1947-48**. A "firka" was an administrative unit consisting of a group of villages. The scheme selected a firka as a unit and tried to develop it fully through agriculture, health, education, cooperatives and rural industries. Village-level workers were appointed to work with the people. It was one of the first government-sponsored attempts at integrated rural development and directly influenced the later Community Development Programme.

3.2.4 Gurgaon Experiment (Punjab / Haryana)

The Gurgaon Experiment was carried out in the Gurgaon district (then in Punjab, now in Haryana) in the 1920s by **Mr. F. L. Brayne**, the Deputy Commissioner. Brayne tried to improve village life through a "village guide" and a programme of sanitation, education, agriculture and removal of social evils. Though it was not fully successful because it was more official and less people-centred, it taught an important lesson: rural development succeeds only when the people themselves want it and take part in it.

3.3 Comparison of pre-independence extension efforts

Effort	Year (approx.)	Place	Founder / Agency	Main idea or contribution
Sriniketan	1921	West Bengal	Rabindranath Tagore	All-round village reconstruction through education and self-reliance

Effort	Year (approx.)	Place	Founder / Agency	Main idea or contribution
Marthandam	1921	Kanyakumari, Tamil Nadu	Dr. Spencer Hatch	"Self-help with the aid of a friendly hand"; three-fold programme
Firka Development Scheme	1947-48	Madras Presidency	Madras Government	Firka as a unit; integrated development through village-level workers
Gurgaon Experiment	1920s	Gurgaon (Punjab/Haryana)	F. L. Brayne	First official rural uplift experiment; lesson that people's will is essential

3.4 Post-independence extension efforts

After independence in 1947, the Government of India launched several organised extension experiments to rebuild rural India. The two most famous are the Etawah Pilot Project and the Nilokheri Experiment. These led directly to the Community Development Programme (1952) and the National Extension Service (1953).

3.4.1 Etawah Pilot Project (Uttar Pradesh)

The Etawah Pilot Project was started in **1948 in Etawah district, Uttar Pradesh**, under the leadership of **Mr. Albert Mayer**, an American engineer-planner, with the help of the Indian administrator **Mr. S. K. Dey**. The project worked with 64 villages to increase agricultural production and improve village life. Its great contribution was the demonstration of a **method of rural extension work** - the "extension approach" of working with the people through village-level workers, using demonstrations, and building the confidence of the farmer. It became the model for the Community Development Programme.

3.4.2 Nilokheri Experiment (Punjab / Haryana)

The Nilokheri Experiment was started in **1948 at Nilokheri in Karnal district (then Punjab, now Haryana)**, under **Mr. S. K. Dey**. Nilokheri was a township built for displaced persons (refugees) after partition. It was a "**self-sufficient township**" combining agriculture, industry, housing, education and health. It showed that rural rehabilitation and integrated development could be achieved by providing employment and self-help opportunities. Nilokheri also strongly influenced the Community Development Programme.

3.4.3 Community Development Programme and National Extension Service

- The **Community Development Programme (CDP)** was launched on **2 October 1952** to promote the all-round development of rural areas with the active participation of the people. It covered the whole country in stages.
- The **National Extension Service (NES)** was started in **1953** to provide the extension machinery at the block level and to extend the CDP to all areas.
- These programmes introduced the **Block Development Officer (BDO)** and the **Village Level Worker (VLW)** - the multi-purpose worker who became the foundation of Indian extension.

3.5 Reorganised Extension System (Training and Visit system)

The Reorganised Extension System, popularly called the **Training and Visit (T&V) system**, was introduced in India in the mid-1970s (around 1974-75) with the support of the **World Bank**, following the ideas of **Dr. Daniel Benor**. It reorganised the whole agricultural extension machinery to make it more effective.

3.5.1 Main features of the T&V system

- **Single line of command:** The extension worker received orders from only one line of authority - the agriculture department - instead of many departments.
- **Regular training:** Extension workers were given regular, fortnightly (every two weeks) training by subject-matter specialists, and the workers then passed the message to farmers.
- **Fixed visits:** Each Village Extension Worker (VEW) had a fixed schedule of visits to a fixed group of contact farmers and other farmers.
- **Contact farmers:** A small number of "contact farmers" (progressive farmers) in each village were used as the first point of contact and as demonstrators.
- **Field-and-farm orientation:** The whole system was oriented to the field and the farm, with work concentrated on the main crops.
- **Time-bound work:** The work was planned on a fortnightly cycle - training, visit, report.
- **Linkage with research:** Subject-matter specialists linked the field with the research station, keeping the technology up to date.

3.5.2 Advantages and limitations

Advantages	Limitations
Regular training kept workers up to date	Too much emphasis on a few contact farmers; other farmers were neglected
Single line of command removed confusion	Concentrated only on crop production; ignored home, health and livestock
Close research-extension linkage	Costly; required many trained staff
Fixed schedule made work systematic	Rigid and mechanical; less attention to the real needs of farmers
Better technology transfer and higher yields	Poor farmers and women were often left out

In Tamil Nadu, the T&V system was implemented through the Department of Agriculture, with Agricultural Officers and Assistant Agricultural Officers as the key field functionaries. Its rigid, top-down character later led to reforms such as ATMA and decentralised, farmer-led extension.

3.6 Timeline of extension efforts

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1920s   Gurgaon Experiment (Brayne)
1921    Sriniketan (Tagore)  and  Marthandam (Spencer Hatch)
1947-48 Firka Development Scheme (Madras)
1948    Etawah Pilot Project  and  Nilokheri Experiment
1952    Community Development Programme (2 October)
1953    National Extension Service
1960s   Intensive Agricultural District / Area Programmes
1974-75 Reorganised Extension System (T&V, with World Bank support)
1974    Krishi Vigyan Kendra (first KVK)
1990s+  ATMA, decentralised and market-led extension, e-extension
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Exam Focus

Key points to remember

- Pre-independence efforts: Sriniketan (Tagore, 1921), Marthandam (Spencer Hatch, 1921), Firka Development Scheme (Madras, 1947-48), Gurgaon Experiment (Brayne, 1920s).
- Marthandam motto: "Self-help with the aid of a friendly hand"; three-fold programme of body, mind and spirit.
- Post-independence: Etawah Pilot Project (Albert Mayer, 1948) and Nilokheri Experiment (S. K. Dey, 1948).
- CDP launched 2 October 1952; NES in 1953; introduced BDO and VLW.
- T&V (Reorganised Extension System) introduced around 1974-75 with World Bank support, based on Benor's ideas.
- T&V features: single line of command, regular fortnightly training, fixed visits, contact farmers, research-extension linkage.
- T&V limits: neglected non-contact farmers, only crop production, costly and rigid.

Short-answer questions with model answers

Q1. Name the four main pre-independence extension efforts in India.

Ans. Sriniketan (1921), Marthandam (1921), the Firka Development Scheme (1947-48) and the Gurgaon Experiment (1920s).

Q2. What was the motto of the Marthandam project and who started it?

Ans. "Self-help with the aid of a friendly hand." It was started in 1921 by Dr. Spencer Hatch at Marthandam (Kanyakumari) with a three-fold programme of body, mind and spirit.

Q3. Expand T&V and state its purpose.

Ans. Training and Visit system (Reorganised Extension System), introduced around 1974-75 with World Bank support to reorganise agricultural extension through regular training of workers and fixed visits to contact farmers.

Q4. Write two limitations of the T&V system.

Ans. (i) It concentrated on a few contact farmers and neglected the rest; (ii) it focused only on crop production and ignored home, health and livestock aspects; it was also costly and rigid.

Long-answer questions with model answer outlines

Q1. Describe the important extension efforts of the pre-independence era in India and their contribution to extension education.

Outline. Explain each of Sriniketan, Marthandam, Firka Development Scheme and Gurgaon Experiment - founder, year, place, main features and contribution. Use the comparison table. Conclude that these experiments established the principles of people's participation, self-help and integrated village development, which shaped the later CDP.

Q2. Explain the Reorganised Extension System (T&V). Discuss its features, advantages and limitations.

Outline. Define T&V and state that it began around 1974-75 with World Bank support on Benor's model. Features: single line of command, regular fortnightly training, fixed visits, contact farmers, field orientation, research linkage. Advantages and limitations from the table. Conclude with its effect in Tamil Nadu and the later move to ATMA and farmer-led extension.

Chapter 4. Extension and Agriculture Development Programmes

4.1 Introduction

After independence, the Government of India and the Indian Council of Agricultural Research (ICAR) launched a series of programmes to increase food production and to reduce rural poverty. These programmes can be grouped as (i) technology-transfer and production programmes, (ii) social justice and poverty alleviation programmes, and (iii) women development programmes. The full forms of the acronyms must be learnt by heart.

4.2 Technology and production programmes

4.2.1 Intensive Agricultural District Programme (IADP)

The IADP, also called the **"Package Programme"**, was launched in **1960-61** in selected districts with the help of the Ford Foundation. Its idea was to concentrate all inputs and services - improved seed, fertiliser, water, credit, machinery and extension - in one "package" in a few districts so that the whole district became a demonstration of improved agriculture. Its success led to the area approach in later programmes.

4.2.2 Intensive Agricultural Area Programme (IAAP)

The IAAP was launched in **1964** to extend the "package" approach to more districts and to larger areas. It aimed at intensive development of selected crops in selected areas, with assured inputs and extension support.

4.2.3 High Yielding Varieties Programme (HYVP)

The HYVP was launched in **1966-67** as part of the "Green Revolution". It promoted high-yielding varieties of wheat, rice, maize, jowar, bajra and millets, along with the package of practices - fertiliser, water and plant protection. It brought about a large increase in foodgrain production and made India self-sufficient in food. Its success depended on irrigation and on the timely supply of inputs.

4.2.4 Krishi Vigyan Kendra (KVK)

The **Krishi Vigyan Kendra** (Farm Science Centre) is a district-level institution of ICAR, the first of which was started in **1974**. Its main purpose is **"vocational training"** for practising farmers, farm women and rural youth, to update their knowledge and skills in agriculture and allied fields. A KVK works on the principle of "**learning by doing**" - the training is mostly practical, on the farm and in the laboratory. A KVK has a **mandatory** programme of training, frontline demonstrations, on-farm testing and advisory services. Tamil Nadu has KVKs in most districts, many of them under TNAU.

4.2.5 Institute Village Linkage Programme (IVLP)

The IVLP was launched by ICAR in **1979** (from an earlier experiment) to link the research institute directly with the villages. Under it, each institute adopted a few villages, diagnosed the problems and tested, refined and transferred technology with the farmers. It promoted the "**participatory**" approach and the idea that research should be farmer-oriented.

4.2.6 Operational Research Project (ORP)

The ORP was started by ICAR in the mid-1970s to test and demonstrate proven technologies in a "cluster" of villages on a watershed or village basis, and to study the problems of adoption. It was a research-cum-extension project that fed its findings back to the research system.

4.2.7 National Demonstration (ND)

The National Demonstration programme, started in the mid-1970s, organised demonstrations of the best available technology on farmers' fields to show the "**yield gap**" - the difference between what the farmer got and what he could get. It was a powerful teaching tool because farmers believe what they see.

4.2.8 National Agricultural Technology Project (NATP)

The NATP was a World Bank-aided project of ICAR launched in the late 1990s. It strengthened research and extension through innovations such as **ATMA** (Agricultural Technology Management Agency) at the district level and competitive research grants. It promoted decentralised, farmer-oriented and market-linked extension.

4.2.9 National Agricultural Innovation Project (NAIP)

The NAIP was a World Bank-aided project of ICAR launched in the mid-2000s to accelerate the "**innovation**" process in agriculture by building partnerships among research, extension, farmers, industry and civil society. Its goal was to convert research results into innovations that reach farmers and improve their livelihoods.

4.3 Summary table of ICAR / Government technology programmes

Programme	Full form	Year (approx.)	Main purpose
IADP	Intensive Agricultural District Programme	1960-61	Package of inputs and services in selected districts
IAAP	Intensive Agricultural Area Programme	1964	Extend package approach to more areas
HYVP	High Yielding Varieties Programme	1966-67	Green Revolution; promote HYV seeds with package of practices
KVK	Krishi Vigyan Kendra	1974	Vocational training, FLD, on-farm testing, advisory
IVLP	Institute Village Linkage Programme	1979	Link research institutes directly with villages
ORP	Operational Research Project	1970s	Test and demonstrate proven technology in village clusters
ND	National Demonstration	1970s	Show the yield gap on farmers' fields
NATP	National Agricultural Technology Project	Late 1990s	Strengthen research and extension; ATMA
NAIP	National Agricultural Innovation Project	Mid-2000s	Build partnerships to convert research into innovation

4.4 Social justice and poverty alleviation programmes

4.4.1 Integrated Tribal Development Agency (ITDA)

The ITDA was created as part of the **Tribal Sub-Plan** approach to develop tribal (Adivasi) areas. Each ITDA covers a tribal development block or project area and implements programmes for tribal families - land development, agriculture, horticulture, education, health and employment. It aims at the **integrated** development of tribal people and the removal of exploitation.

4.4.2 Integrated Rural Development Programme (IRDP)

The IRDP was launched in **1978-79** to help the poorest rural families cross the "**poverty line**" by giving them productive assets (milch animals, sheep, sewing machines, small tools) and subsidy-linked bank credit, so that the assets would generate a regular income. It was one of the largest poverty alleviation programmes and was later merged into the Swarnjayanti Gram Swarozgar Yojana (SGSY).

4.4.3 Swarnjayanti Gram Swarozgar Yojana (SGSY)

The SGSY was launched in **1999** by merging several self-employment programmes (including the IRDP). It aimed at bringing poor families above the poverty line by organising them into **Self-Help Groups (SHGs)** and giving them training, credit and subsidy for self-employment activities. The SGSY emphasised the group approach and was later restructured as the National Rural Livelihoods Mission (NRLM).

4.4.4 National Rural Livelihoods Mission (NRLM) / Aajeevika

The NRLM, also called **Aajeevika**, was launched in **2011** to reduce rural poverty by promoting diversified livelihoods and strong grassroots institutions, especially **women's Self-Help Groups**. Its approach is "**demand-driven**" - the poor decide their own livelihood activities and the mission provides the support. In Tamil Nadu the mission is implemented through the **Tamil Nadu Rural Livelihoods Mission (TNRTP / TNRLM)** with a large network of women's SHGs.

4.5 Women development programmes

4.5.1 Rashtriya Mahila Kosh (RMK)

The RMK (National Credit Fund for Women) was set up in **1993** as a society under the Ministry of Women and Child Development. It provides **micro-credit** to poor women through intermediate organisations (NGOs and SHGs) so that women can take up income-generating activities. Its aim is to empower women economically.

4.5.2 Mahila Samridhi Yojana (MSY)

The Mahila Samridhi Yojana, launched in **1993**, encouraged poor rural women to save money in a **post office savings account** by paying a small incentive on their deposits. Its purpose was to develop the **habit of thrift and saving** among rural women and to make them financially self-reliant.

4.6 Summary table of social justice, poverty and women programmes

Programme	Full form	Focus	Main feature
ITDA	Integrated Tribal Development Agency	Tribal development	Integrated development of tribal areas
IRDP	Integrated Rural Development Programme	Poverty alleviation	Productive assets with subsidy-linked credit
SGSY	Swarnjayanti Gram Swarozgar Yojana	Self-employment	SHG-based self-employment
NRLM	National Rural Livelihoods Mission	Rural livelihoods	Demand-driven; women's SHGs (Aajeevika)
RMK	Rashtriya Mahila Kosh	Women	Micro-credit to poor women
MSY	Mahila Samridhi Yojana	Women	Savings habit through post office accounts

Exam Focus

Key points to remember

- IADP (1960-61) = Package Programme; IAAP (1964); HYVP (1966-67) = Green Revolution.
- KVK (1974) = Farm Science Centre; vocational training, learning by doing, FLD.
- IVLP (1979), ORP (1970s), ND (1970s) - research-extension linkage and demonstrations.
- NATP (late 1990s) introduced ATMA; NAIP (mid-2000s) built innovation partnerships.
- Poverty alleviation: IRDP (1978-79), SGSY (1999), NRLM / Aajeevika (2011) with women's SHGs.
- ITDA = Integrated Tribal Development Agency.
- Women programmes: RMK (1993, micro-credit) and MSY (1993, savings habit).

Short-answer questions with model answers

Q1. Expand IADP, IAAP and HYVP and give their years.

Ans. IADP - Intensive Agricultural District Programme (1960-61); IAAP - Intensive Agricultural Area Programme (1964); HYVP - High Yielding Varieties Programme (1966-67).

Q2. What is the main objective of a Krishi Vigyan Kendra?

Ans. To provide vocational training in agriculture and allied fields to practising farmers, farm women and rural youth, based on the principle of learning by doing, along with frontline demonstrations, on-farm testing and advisory services.

Q3. Expand SGSY and NRLM.

Ans. SGSY - Swarnjayanti Gram Swarozgar Yojana; NRLM - National Rural Livelihoods Mission (also called Aajeevika).

Q4. Name any two women development programmes.

Ans. Rashtriya Mahila Kosh (RMK, 1993) and Mahila Samridhi Yojana (MSY, 1993).

Long-answer questions with model answer outlines

Q1. Describe the important agriculture development programmes launched by ICAR / Government of India from IADP to NAIP.

Outline. Introduce the need for such programmes. Explain each - IADP, IAAP, HYVP, KVK, IVLP, ORP, ND, NATP, NAIP - with year, purpose and key feature. Use the summary table. Conclude that the trend has moved from top-down package programmes to participatory, farmer-oriented and innovation-based approaches.

Q2. Explain the social justice, poverty alleviation and women development programmes of the Government of India.

Outline. Group them: social justice and poverty (ITDA, IRDP, SGSY, NRLM); women (RMK, MSY). Explain each with year and feature. Discuss the shift from individual-asset to group and demand-driven approaches (SHGs). Conclude with the role of extension in implementing these programmes at the village level.

Chapter 5. New Trends in Agriculture Extension

5.1 Why extension is changing

The traditional public extension system was built for the Green Revolution era, when the main job was to move a standard package of practices from the research station to the farmer. Today the situation has changed: farms are more market-oriented, crops are more diverse, farmers are more educated, information is available from many sources, and the public extension staff are too few to reach every farmer. This has produced new trends in extension, of which the most important are **privatisation of extension** and **cyber extension / e-extension**.

5.2 Privatisation of extension

5.2.1 Meaning and definition

Privatisation of extension means the transfer of extension services from the government (public sector) to private agencies, or the provision of extension services by private bodies on a payment basis. The private agency may be a company, a cooperative, an NGO, an agri-business firm or an input dealer.

In simple terms, **privatised extension is extension service provided by a private organisation, often for a fee or as a support to the sale of a product**.

5.2.2 Forms of private extension

- **Input-dealer extension:** Dealers of seeds, fertilisers and pesticides advise farmers, often as part of selling their products.
- **Contract-farming extension:** Companies that buy the crop (e.g. for sugar, cotton, vegetables, dairying) provide advice and inputs to their contract farmers.
- **NGO extension:** Voluntary organisations provide extension to their target groups, often for the poor and for women.
- **Fee-based advisory:** Private consultants, agri-clinics and agri-business centres provide paid advice.
- **Cooperative extension:** Dairy, sugar and other cooperatives advise their members.

5.2.3 Advantages and limitations

Advantages of privatised extension	Limitations of privatised extension
More staff and wider reach; not limited by government posts	Only farmers who can pay or who buy a product are served
Quick, timely and specialised advice	Small and marginal farmers and the poor may be left out
Better for commercial and high-value crops	Advice may be biased towards the company's products
Cost shared by the user, reducing the public burden	No single agency for the whole village; duplication possible
Strong market linkage for the farmer	Weak in remote areas where profit is low

5.3 Cyber extension / e-extension

5.3.1 Meaning

Cyber extension, or e-extension, means the use of **information and communication technology (ICT)** - the internet, computers, mobile phones, satellites, radio, television and multimedia - to deliver extension information and advisory services to farmers. It makes extension faster, wider and two-way, because the farmer can also send back his questions and problems.

5.3.2 Components and tools of cyber extension

- **Web portals and websites:** Agricultural information portals run by ICAR, TNAU, the Department of Agriculture and others.
- **Mobile applications:** The **Uzhavan app** of Tamil Nadu is a single-window app giving crop advisories, weather, market prices, machinery booking and subsidy information. The **m-Velanmai** app is a mobile version of the e-Velanmai advisory service.
- **Video conferencing:** Experts at a distant research station talk to farmers in a village tele-centre; the farmer can see and question the expert. **e-Velanmai** uses this participatory approach.
- **Podcasts and audio/video:** Recorded audio and video lessons that farmers can listen to or watch at their convenience.
- **Multimedia:** The combined use of text, sound, pictures, animation and video in a single learning package; makes learning easier for low-literacy learners.
- **Community radio and television:** Local radio and TV programmes on agriculture.
- **Call centres and help lines:** Kisan Call Centres give free advice on a toll-free number.
- **Social media:** WhatsApp groups, YouTube channels and Facebook pages for quick sharing of advisories.

5.3.3 e-Velanmai and m-Velanmai (Tamil Nadu)

- **e-Velanmai** ("electronic farming") is an ICT-based, **participatory extension** system developed in Tamil Nadu (with TNAU and the M. S. Swaminathan Research Foundation). In each participating village, a trained **"e-Velanmai farmer"** collects the problems and crop samples of fellow farmers, and through a computer and internet connection forwards queries to a panel of scientists. The scientists reply with advice, which is passed back to the farmers. It makes the advisory **two-way and farmer-driven**.
- **m-Velanmai** is the **mobile application** version, allowing farmers to send queries, photographs and get expert advice directly through a smartphone, without waiting for a computer centre.

5.3.4 Advantages and limitations of cyber extension

Advantages	Limitations
Fast, wide and low-cost reach	Needs connectivity, electricity and a device
Two-way: farmers can ask questions	Digital divide - the poor and the elderly may be left out
Multimedia suits low-literacy learners	Content may be in a language or format not useful locally
Timely weather, price and pest alerts	Trust and local relevance of advice must be built
Records of queries help in planning	Requires trained staff and maintenance

5.4 Multimedia in extension

Multimedia is the integration of two or more media - text, graphics, audio, video and animation - in one package. In extension, multimedia is used in CD/DVD lessons, mobile apps, e-learning modules and interactive kiosks. It is useful because:

- It caters to different learning styles (visual, auditory, reading).
- It reduces dependence on literacy; pictures and audio help illiterate farmers.
- It allows the learner to repeat the lesson any number of times.
- It is standard, so every learner receives the same message.
- It can be updated easily and distributed widely at low cost.

5.5 Worked example - choosing the right new-extension tool

Problem. A district has (a) a group of literate, smartphone-using commercial vegetable farmers near a town, and (b) a remote tribal hamlet with poor connectivity and low literacy. Suggest a suitable extension approach for each.

Solution.

Situation	Suitable approach	Reason
(a) Literate commercial farmers	Mobile app (Uzhavan), WhatsApp advisories, e-Velanmai, fee-based private advisory	They can use ICT, need timely market and price information
(b) Remote tribal hamlet	Community radio, group meetings, farm and home visits, local leader, printed pictorial material	Poor connectivity and low literacy; needs face-to-face and audio/pictorial methods

The lesson is that new tools do not replace the old ones; they are added to the extension "media mix" and chosen according to the situation.

Exam Focus

Key points to remember

- Privatisation of extension = extension provided by private agencies, often for a fee or linked to a product.
- Forms: input-dealer, contract-farming, NGO, fee-based and cooperative extension.
- Cyber extension / e-extension = use of ICT to deliver advisory services to farmers.
- Tools: portals, mobile apps, video conferencing, podcasts, multimedia, community radio, call centres, social media.
- e-Velanmai = participatory ICT advisory with an e-Velanmai farmer and scientists; m-Velanmai = its mobile app.
- Uzhavan app (Tamil Nadu) = single-window app for advisories, weather, price, machinery and subsidy.
- New tools are added to the media mix, not substituted for older methods.

Short-answer questions with model answers

Q1. What is privatisation of extension?

Ans. Privatisation of extension means the provision of extension services by private agencies (companies, NGOs, cooperatives, consultants), often for a fee or as support to the sale of a product, instead of by the government.

Q2. Define cyber extension.

Ans. Cyber extension (e-extension) is the use of information and communication technology - internet, mobile phones, video conferencing, multimedia - to deliver extension information and advisory services to farmers in a fast, wide and two-way manner.

Q3. What is e-Velanmai?

Ans. e-Velanmai is an ICT-based participatory extension system of Tamil Nadu in which a trained e-Velanmai farmer collects fellow farmers' problems and sends them to a panel of scientists through the internet, and passes back the advice received.

Q4. Name any four tools of cyber extension.

Ans. Web portals, mobile apps (Uzhavan, m-Velanmai), video conferencing and podcasts; also multimedia, community radio, call centres and social media.

Long-answer questions with model answer outlines

Q1. Discuss the new trends in agricultural extension with special reference to privatisation and cyber extension.

Outline. Introduce why extension is changing. Explain privatisation - meaning, forms, advantages and limitations (table). Explain cyber extension - meaning, tools, e-Velanmai and m-Velanmai, Uzhavan app, advantages and limitations. Mention multimedia. Conclude with the media-mix idea and the digital divide as a challenge.

Q2. What is e-extension? Explain its tools and importance with Tamil Nadu examples.

Outline. Define cyber/e-extension. List the tools - portals, apps, video conferencing, podcasts, multimedia, radio, call centres, social media. Explain e-Velanmai and m-Velanmai and the Uzhavan app. Give advantages (speed, reach, two-way, low cost) and limitations (connectivity, digital divide). Conclude that e-extension complements, not replaces, traditional extension.

Chapter 6. Market-led and Farmer-led Extension; Expert Systems

6.1 Market-led extension

6.1.1 Meaning and definition

Market-led extension is an approach in which extension advice is guided by the **market** - that is, by what the consumer wants and what will fetch a good price - rather than only by production. Instead of asking only "how can I grow more?", the farmer and the extension worker ask "what should I grow, of what quality, and where can I sell it profitably?"

Definition: Market-led extension is a demand-driven extension approach that helps farmers to produce according to market demand and to market their produce profitably, by linking production with processing, value addition and the market.

6.1.2 Features and importance

- **Demand-driven, not supply-driven:** Production is planned from the market backward to the farm.
- **Focus on quality:** The market pays for quality; extension teaches grading, packaging and post-harvest handling.
- **Value addition and processing:** Extension helps farmers to add value (cleaning, grading, processing) to get a higher price.
- **Market information:** Extension provides price and demand information so the farmer can decide what and when to sell.
- **Linkages:** It builds links among farmers, traders, processors, exporters and consumers.
- **Farmer organisations:** It works through Farmer Producer Organisations (FPOs) so that small farmers can bargain as a group.
- **Risk reduction:** By matching production to demand, it reduces the risk of surplus and low prices.

Importance: Market-led extension raises farm income, encourages diversification to high-value crops, reduces the distress of price crashes, promotes quality and exports, and makes extension more relevant to the farmer's real problem - which is income, not merely yield.

6.1.3 Example from Tamil Nadu

Tamil Nadu promotes market-led extension through **Uzhavan app's market price module**, through **regulated markets (uzhavar sandhai - farmers' markets)** where farmers sell directly to consumers, and through **FPOs** for crops such as banana, coconut, vegetables and flowers. Extension workers help farmers plan crops according to the price trend in the nearest regulated market.

6.2 Farmer-led extension

6.2.1 Meaning and definition

Farmer-led extension is an approach in which **farmers themselves** - individually or through their organisations - take the lead in planning, organising and delivering extension to fellow farmers. The farmer is not a passive receiver of advice but an active teacher, planner and decision-maker. The classic example is the **Farmer Field School (FFS)**, where farmers learn by observing and experimenting in their own fields and then teach one another.

Definition: Farmer-led extension is a participatory approach in which farmers, often organised into groups, take the initiative and responsibility for generating and sharing agricultural knowledge among themselves, with the extension agency acting as a facilitator.

6.2.2 Features and importance

- **Farmer as the teacher:** Knowledge flows horizontally, farmer to farmer, not only vertically.
- **Local relevance:** Advice is tested and adapted in the local field, so it is more acceptable.
- **Empowerment:** Farmers gain confidence, decision-making power and leadership.
- **Lower cost:** It uses existing local knowledge and reduces the burden on paid staff.
- **Sustainability:** Once trained, farmers can continue without external help.
- **Participation:** It fits the principle of participation and the principle of leadership.

Importance: Farmer-led extension is important because farmers trust other farmers more than they trust officials, because it reaches more farmers at low cost, and because it builds the capacity of the community to solve its own problems. It is the practical form of the "farmer-back-to-farmer" idea.

6.2.3 Market-led versus farmer-led extension

Point	Market-led extension	Farmer-led extension
Driving force	Market demand and price	The farmers themselves
Main aim	Profitable production and marketing	Farmers teaching and organising themselves
Who leads	Extension + market actors	Farmers and their groups
Key tool	Market information, FPO, value chain	Farmer Field School, farmer-to-farmer learning
Focus	Quality, price, value addition	Participation, empowerment, local knowledge
Example	Uzhavan price module, uzhavar sandhai	FFS, farmer trainers, e-Velanmai farmer

6.3 Expert systems

6.3.1 Meaning and definition

An expert system is a **computer program** that uses stored knowledge and a set of rules to give advice or solve problems in a narrow field, in the way a human expert would. It "captures" the knowledge of one or more human experts in a particular subject and makes it available to many users through a computer or mobile phone.

Definition: An expert system is a computer-based information system that imitates the decision-making ability of a human expert by using a knowledge base and an inference engine to give reasoned advice to the user.

6.3.2 Components of an expert system

- **Knowledge base:** The facts and rules collected from human experts (for example, rules linking symptoms to diseases).
- **Inference engine:** The reasoning part that applies the rules to the facts given by the user and arrives at a conclusion.
- **User interface:** The part through which the farmer asks questions and receives the advice.

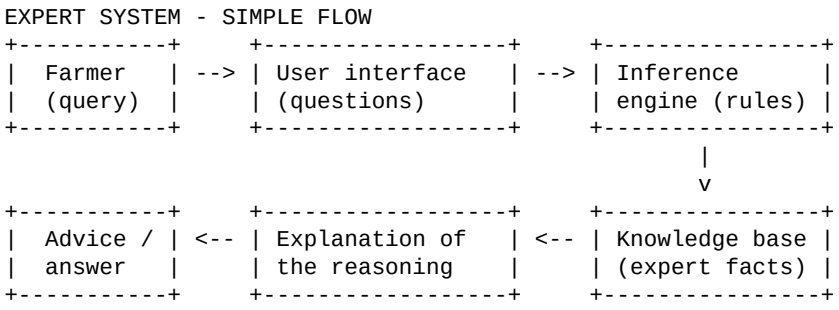
- **Explanation facility:** The part that explains how the conclusion was reached.

6.3.3 Applications in agriculture

- **Crop disease diagnosis:** The farmer enters the crop, the symptoms and the weather, and the system suggests the likely disease and its control (for example, a paddy pest diagnosis system).
- **Fertiliser recommendation:** The system advises the correct dose of fertiliser from soil-test values and the crop.
- **Irrigation scheduling:** Advice on when and how much to irrigate.
- **Weather-based advisories:** Combining weather data with crop rules to warn of pest and disease outbreaks.
- **Livestock health:** Diagnosis of animal diseases and advice on treatment.
- **Farm planning and marketing:** Advice on crop choice, cost of cultivation and marketing.

6.3.4 Advantages and limitations

Advantages	Limitations
Makes expert advice available anywhere, at any time	Needs a large, accurate knowledge base
Consistent advice; does not get tired	Cannot handle situations outside its rules
Fast and low-cost per query	Cannot truly "understand" like a human expert
Can be delivered through mobile apps	Needs a device, power and connectivity
Frees human experts for difficult cases	Knowledge must be updated regularly



Exam Focus

Key points to remember

- Market-led extension = demand-driven extension guided by the market; focuses on what to grow, quality and profitable selling.
- Features: demand-driven, quality, value addition, market information, FPOs, risk reduction.
- Farmer-led extension = farmers lead in generating and sharing knowledge; e.g. Farmer Field School.
- Market-led (market-driven) differs from farmer-led (farmer-driven) - see comparison table.
- Expert system = computer program using a knowledge base and inference engine to give expert-like advice.
- Components: knowledge base, inference engine, user interface, explanation facility.

- Applications: disease diagnosis, fertiliser and irrigation advice, weather-based advisories, livestock health, marketing.

Short-answer questions with model answers

Q1. Define market-led extension.

Ans. Market-led extension is a demand-driven approach that helps farmers produce according to market demand and market their produce profitably, linking production with value addition and the market.

Q2. What is farmer-led extension?

Ans. Farmer-led extension is a participatory approach in which farmers themselves, often through groups such as Farmer Field Schools, take the lead in generating and sharing agricultural knowledge, with the extension agency acting as a facilitator.

Q3. Define an expert system.

Ans. An expert system is a computer-based system that imitates the decision-making of a human expert using a knowledge base and an inference engine to give reasoned advice.

Q4. Name the components of an expert system.

Ans. Knowledge base, inference engine, user interface and explanation facility.

Long-answer questions with model answer outlines

Q1. Explain the meaning, features and importance of market-led and farmer-led extension with examples.

Outline. Define both. Give features of each. Present the comparison table. Examples: Uzhavan price module and uzhavar sandhai for market-led; Farmer Field School and e-Velanmai farmer for farmer-led. Discuss importance - income, quality, empowerment, low cost. Conclude that both shift the farmer from a passive receiver to an active partner.

Q2. What is an expert system? Describe its components and applications in agriculture.

Outline. Define expert system. Explain the four components with the flow diagram. List applications - disease diagnosis, fertiliser recommendation, irrigation, weather-based advisory, livestock health, marketing. Give advantages and limitations from the table. Conclude with the role of expert systems in e-extension.

Chapter 7. Rural Development

7.1 Concept and meaning of rural development

Rural development means the process of improving the quality of life and the economic well-being of people living in rural areas. It is not limited to agriculture; it includes the development of rural infrastructure (roads, water, electricity, housing), rural institutions (panchayats, cooperatives, SHGs), health, education, employment and social justice.

The concept of rural development rests on three ideas:

- **Change:** It is a process of change from a poorer to a better condition of life.
- **Comprehensiveness:** It covers all aspects of rural life - economic, social, cultural and political.
- **Participation:** It must be carried out with and by the rural people, not imposed on them.

7.2 Definitions of rural development

- **World Bank:** Rural development is a strategy designed to improve the economic and social life of a specific group of people - the rural poor.
- **Robert Chambers:** Rural development is a strategy to enable a specific group of people - poor rural men and women - to gain for themselves and their children more of what they want and need.
- **Singh:** Rural development is the process of improving the standard of living of the rural people through the provision of basic amenities and opportunities.
- **A. T. Mosher:** Rural development is the development of the rural people - it means raising their incomes, their health, their education and their participation in the life of the community.

The common thread in these definitions is that rural development is **for the rural poor**, is **multi-dimensional**, and requires the **participation** of the people.

7.3 Objectives and importance of rural development

- To reduce poverty and unemployment in rural areas.
- To raise rural incomes and the standard of living.
- To provide basic amenities - roads, drinking water, housing, electricity, health and education.
- To develop rural infrastructure and rural institutions.
- To reduce migration from villages to cities.
- To bring about social justice and remove inequality.
- To develop the skills, leadership and self-reliance of rural people.

7.4 Rural development programmes of the Government of India

The Government of India has launched a large number of rural development programmes. The important ones named in the syllabus are DWCRA, MGNREGA, RSVY, IAY, SGRY, ITDP and PMGSY.

7.4.1 Development of Women and Children in Rural Areas (DWCRA)

The DWCRA was launched in **1982-83** as a sub-scheme of IRDP. Its purpose was to improve the income and status of **rural women** by organising them into groups and helping them take up income-generating activities such as dairying, tailoring, handicrafts and small trade. The DWCRA group is one of the earliest forms of the **women's Self-Help Group** in India.

7.4.2 Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA)

The MGNREGA, originally the National Rural Employment Guarantee Act (NREGA) of **2005**, provides a **legal guarantee of at least 100 days of unskilled manual wage employment** in a financial year to every rural household whose adult members are willing to do public work. The work is mostly on rural infrastructure - water conservation, irrigation, roads and land development. Wages are paid through bank or post office accounts. Its aims are to provide a livelihood safety net, create durable rural assets and reduce distress migration.

7.4.3 Rashtriya Sam Vikas Yojana (RSVY)

The RSVY was launched in **2003-04** to develop the **backward districts** of the country. It provided funds to selected backward districts for programmes that would remove the causes of backwardness and speed up development - agriculture, irrigation, rural roads, education and health. It is also called the Backward Districts Initiative.

7.4.4 Indira Awaas Yojana (IAY)

The IAY was launched in **1985-86** as part of the rural employment programmes and later became a major **rural housing** scheme. It provides financial assistance to build houses for **below-poverty-line (BPL)** rural families, especially SC/ST families and freed bonded labourers. Its aim is to provide a safe and durable shelter, now continued through the Pradhan Mantri Awaas Yojana - Gramin (PMAY-G).

7.4.5 Sampoorna Grameen Rozgar Yojana (SGRY)

The SGRY was launched in **2001** by merging the Employment Assurance Scheme and the Jawahar Gram Samridhi Yojana. It aimed to provide **additional wage employment** and food security in rural areas, along with the creation of durable community assets, irrigation works and rural infrastructure. It was later merged into the MGNREGA.

7.4.6 Integrated Tribal Development Project (ITDP)

The ITDP is the **project-level unit for tribal development** under the Tribal Sub-Plan approach. Each ITDP covers a tribal development block and implements programmes for the all-round development of tribal families - land, agriculture, horticulture, education, health, drinking water and employment. It works through the ITDA (Integrated Tribal Development Agency).

7.4.7 Pradhan Mantri Gram Sadak Yojana (PMGSY)

The PMGSY was launched in **2000** to provide **all-weather road connectivity** to unconnected rural habitations. Good rural roads are essential because they link villages to markets, hospitals and schools, reduce transport cost and support the whole rural economy. It is implemented by the National Rural Roads Development Agency with the State Governments.

7.5 Summary table of rural development programmes

Programme	Full form	Year (approx.)	Main purpose
DWCRA	Development of Women and Children in Rural Areas	1982-83	Income and status of rural women through groups

Programme	Full form	Year (approx.)	Main purpose
MGNREGA	Mahatma Gandhi National Rural Employment Guarantee Act	2005	Guarantee 100 days of wage employment per rural household
RSVY	Rashtriya Sam Vikas Yojana	2003-04	Development of backward districts
IAY	Indira Awaas Yojana	1985-86	Housing for BPL rural families
SGRY	Sampoorna Grameen Rozgar Yojana	2001	Wage employment, food security, rural assets
ITDP	Integrated Tribal Development Project	Under Tribal Sub-Plan	All-round development of tribal areas
PMGSY	Pradhan Mantri Gram Sadak Yojana	2000	All-weather road connectivity to villages

7.6 Worked example - matching a programme to a need

Problem. A village has (a) a group of women with no income of their own, (b) an unconnected hamlet cut off in the rains, and (c) several landless labourers without work in the dry season. Match a suitable government programme to each need.

Solution.

Need	Suitable programme	Why
(a) Income for women	DWCRA (now SHG / NRLM)	Organises women into groups for income-generating activities
(b) Road connectivity	PMGSY	Provides all-weather road to unconnected habitations
(c) Work for landless labourers	MGNREGA	Guarantees 100 days of wage employment per household

Exam Focus

Key points to remember

- Rural development = improving the quality of life and economic well-being of rural people; multi-dimensional and participatory.
- Key ideas: change, comprehensiveness, participation; it is for the rural poor.
- DWCRA (1982-83) = women's groups for income; an early form of SHG.
- MGNREGA (2005) = legal guarantee of 100 days of wage employment per rural household.
- RSVY (2003-04) = backward districts; IAY (1985-86) = rural housing for BPL.
- SGRY (2001) = wage employment and rural assets; ITDP = tribal development project.
- PMGSY (2000) = all-weather road connectivity to villages.

Short-answer questions with model answers

Q1. Define rural development.

Ans. Rural development is a strategy and process to improve the economic and social life of the rural poor, covering agriculture, infrastructure, institutions, health, education and employment, with the participation of the people.

Q2. What is MGNREGA and what does it guarantee?

Ans. The Mahatma Gandhi National Rural Employment Guarantee Act (2005) guarantees at least 100 days of unskilled manual wage employment in a financial year to every rural household whose adult members are willing to work.

Q3. Expand DWCRA and PMGSY.

Ans. DWCRA - Development of Women and Children in Rural Areas; PMGSY - Pradhan Mantri Gram Sadak Yojana.

Q4. State the main aim of the IAY.

Ans. To provide financial assistance for building houses for below-poverty-line rural families, especially SC/ST families, so that they get safe and durable shelter.

Long-answer questions with model answer outlines

Q1. Explain the concept and meaning of rural development and describe the important rural development programmes of the Government of India.

Outline. Define rural development with two definitions; explain the three ideas - change, comprehensiveness, participation. State objectives and importance. Then describe DWCRA, MGNREGA, RSVY, IAY, SGRY, ITDP and PMGSY with year and purpose, using the summary table. Conclude with the shift towards rights-based and participatory programmes.

Q2. Discuss the rural development programmes launched by the Government of India with special reference to employment, housing and connectivity.

Outline. Group the programmes: employment and livelihood (MGNREGA, SGRY, DWCRA), housing (IAY), connectivity and infrastructure (PMGSY), area and tribal development (RSVY, ITDP). Explain each with year and feature. Give a village example matching needs to programmes. Conclude with their role in poverty reduction.

Chapter 8. Mid Semester Examination

8.1 Purpose of the mid semester examination

The mid semester examination of AEX 201 is held after the completion of the first seven lectures. It tests the fundamental concepts of extension education - its meaning, types, principles and scope; programme planning; the history of extension in India; and the major development programmes. It is usually a written test of short duration, and it carries a portion of the internal assessment marks.

8.2 What the mid semester examination covers

- Lecture 1: Education and Extension Education - meaning, definitions, types, scope, process, objectives and principles.
- Lecture 2: Extension Programme Planning - meaning, definition, process, principles and steps.
- Lecture 3: Extension Systems in India - pre- and post-independence efforts and the T&V system.
- Lecture 4: Extension and agriculture development programmes; social justice, poverty alleviation and women development programmes.
- Lectures 5 and 6 (if covered): New trends in extension; market-led and farmer-led extension; expert systems.
- Lecture 7 (if covered): Rural development and its programmes.

8.3 Revision checklist before the mid semester examination

- Can you define education and extension education, and give two definitions of each?
- Can you draw the comparison table of formal, informal and non-formal education?
- Can you list the objectives and principles of extension education?
- Can you draw the programme planning cycle and name all eight steps?
- Can you name the four pre-independence efforts with founders and years?
- Can you explain the T&V system and its advantages and limitations?
- Can you give the full forms and years of IADP, IAAP, HYVP, KVK, IVLP, ORP, ND, NATP and NAIP?
- Can you distinguish market-led from farmer-led extension?
- Can you name the rural development programmes and their purposes?

Tip for the examination: In every answer, first write the definition, then the points, then one Indian or Tamil Nadu example, and finish with a one-line conclusion. This structure earns marks for completeness.

Exam Focus

Key points to remember

- The mid semester examination covers the first seven lectures of the syllabus.
- It usually carries a share of the internal assessment marks.
- Definitions, comparison tables, cycles and full forms of acronyms are the most frequently asked items.

- Always answer with definition, points, example and conclusion.
- Use the revision checklist to identify weak areas.

Short-answer questions with model answers

Q1. Which lectures are covered by the mid semester examination?

Ans. Lectures 1 to 7 - education and extension education, programme planning, extension systems in India, development programmes, new trends, market-led and farmer-led extension and rural development.

Q2. Name any three items you must be able to draw from memory.

Ans. The comparison table of types of education, the programme planning cycle, and the T&V advantages-limitations table (also the extension process cycle).

Long-answer questions with model answer outlines

Q1. How would you prepare for the AEX 201 mid semester examination?

Outline. List the topics of lectures 1 to 7; state that definitions, tables, cycles and acronyms matter most; describe a study plan - read, then revise tables and diagrams, then write model answers; use the revision checklist; practise writing structured answers with examples.

Chapter 9. Community Development

9.1 Meaning of community development

A **community** is a group of people living in a definite geographical area, with common interests and a feeling of belonging, who work together for common goals. A village is a community. **Community development** means the process by which the members of a community come together, plan and act to improve the economic, social and cultural conditions of their whole community, and in doing so develop their own abilities and confidence.

Community development is thus both a **method** (working with the whole community) and a **programme** (a set of activities for improving community life). It is built on the belief that the community itself, with a little help, can solve its own problems.

9.2 Definitions of community development

- **United Nations:** Community development is the process by which the efforts of the people themselves are united with those of governmental authorities to improve the economic, social and cultural conditions of communities, to integrate these communities into the life of the nation, and to enable them to contribute fully to national progress.
- **Community Development Programme of India:** Community development is the process of helping the people to help themselves, so that they may improve their own living conditions and become self-reliant citizens.
- **Sanders:** Community development is a process of change in the community, brought about by the people themselves, aimed at the improvement of the community as a whole.

From these definitions, the essential elements of community development are: **the people's own efforts**, **the help of government or other agencies**, **the improvement of economic, social and cultural conditions**, and **integration with the nation**.

9.3 Concept of community development

The concept of community development rests on the following ideas:

- **The community as a whole is the unit of work** - not an individual, a family or a single crop.
- **Self-help and participation** - the people must plan and work for their own development.
- **Change from within** - change is lasting only when it comes from the people themselves.
- **Balanced development** - all aspects of community life (agriculture, health, education, recreation) must develop together.
- **Democratic method** - decisions are taken by the people, with the worker as a guide and friend.
- **Use of local resources and local leadership** - the community's own resources and leaders are used first.
- **Government as a partner, not a master** - the government provides technical help, funds and materials, but the people provide the effort.

9.4 Principles of community development

- **Principle of the community as a whole:** Development must cover the whole community, not isolated individuals.

- **Principle of participation:** People must take part in every stage - planning, execution and evaluation.
- **Principle of self-help:** The programme should help people to help themselves and become self-reliant.
- **Principle of local leadership:** Use and develop local leaders to lead the work.
- **Principle of felt needs:** Start from the needs that the people themselves feel.
- **Principle of cooperation:** All agencies - official and non-official - must work together.
- **Principle of balanced development:** Economic, social and cultural development must go together.
- **Principle of flexibility:** The programme must be adapted to the local situation.
- **Principle of democratic functioning:** Decisions should be taken democratically, with the people.
- **Principle of continuity:** Development is a continuous process, not a one-time project.

9.5 Philosophy of community development

The philosophy of community development is the belief that **people have within them the capacity to develop themselves**, and that the job of the development worker is to awaken, encourage and support this capacity rather than to do things for the people. The worker's role is that of a **friend, philosopher and guide**.

The philosophy is often expressed in three famous sayings:

- "**Help people to help themselves.**" - the people, not the agency, are the centre of development.
- "**Work with the people, not for the people.**" - participation is the method.
- "**Start where the people are.**" - begin with the people's own situation, needs and level.

The philosophy also has a moral dimension: community development aims at the **all-round development of the human personality** and at building a just and equal society. It respects the dignity of the individual, values democracy, and believes in the unity of the community.

9.6 Community development versus community organisation

Point	Community development	Community organisation
Focus	Development of the whole community - economic, social, cultural	Organisation of the community into groups and institutions
Main aim	Improve living conditions and develop self-reliance	Build and strengthen community institutions
Method	Educational, through participation and self-help	Organising, planning and coordinating group action
Relation	Community organisation is a part of community development	A narrower, institutional part of the wider process
Example	CDP of 1952	Formation of an SHG or a village panchayat committee

9.7 The Community Development Programme in India

The Community Development Programme (CDP) of 1952 (see Chapter 3) was the practical application of the philosophy of community development in India. It covered the whole country in stages, used the block as the unit of administration, and placed the **Village Level Worker (VLW)** at the village level to work with the people. Its aims were to raise agricultural production, improve health and sanitation, develop rural industries and education, and build community feeling.

Exam Focus

Key points to remember

- Community = people in a definite area with common interests; community development = people planning and acting together to improve their whole community.
- UN definition: people's efforts united with government efforts to improve economic, social and cultural conditions.
- Concept: whole community as unit, self-help, participation, balanced development, democratic method, local resources.
- Principles: whole community, participation, self-help, local leadership, felt needs, cooperation, balance, flexibility, democracy, continuity.
- Philosophy: "help people to help themselves"; "work with the people, not for the people"; "start where the people are".
- The worker is a friend, philosopher and guide.
- CDP of 1952 is the practical application of community development in India.

Short-answer questions with model answers

Q1. Define community development.

Ans. Community development is the process by which the people's own efforts are united with those of government or other agencies to improve the economic, social and cultural conditions of the whole community and to make it self-reliant.

Q2. State any four principles of community development.

Ans. (i) The community as a whole is the unit; (ii) participation of the people; (iii) self-help; (iv) use of local leadership. Others: felt needs, cooperation, balanced development, flexibility, democracy and continuity.

Q3. What is the philosophy of community development?

Ans. The belief that people have the capacity to develop themselves, and the worker's role is to awaken and support that capacity. It is summed up as "help people to help themselves", "work with the people, not for the people", and "start where the people are".

Q4. Give the UN definition of community development.

Ans. Community development is the process by which the efforts of the people themselves are united with those of governmental authorities to improve the economic, social and cultural conditions of communities and to integrate them into the life of the nation.

Long-answer questions with model answer outlines

Q1. Explain the meaning, concept, principles and philosophy of community development.

Outline. Define community and community development; give the UN definition. Explain the concept - whole community as unit, self-help, participation, balanced development, democracy, local resources. List and explain the principles. Explain the philosophy with the three sayings and the role of the worker as friend, philosopher and guide. Conclude with the CDP of 1952 as its practical application.

Q2. "Community development is a process of helping people to help themselves." Discuss with reference to principles and philosophy.

Outline. Interpret the statement; show that self-help and participation are the core. Explain the principles and the philosophy. Give examples - a village building its own school with government help, an SHG managing its own savings. Contrast with a top-down approach. Conclude that lasting development is self-development.

Chapter 10. Rural Leadership

10.1 Concept of leadership

Leadership is the quality or the process by which a person influences, guides and inspires other people to work together willingly towards a common goal. A leader is a person who has followers, who can get people to act, and who is accepted by the group as its guide. Leadership is not the same as authority or power that comes from a post; real leadership comes from the consent and trust of the followers.

In the rural context, leadership is the ability of a person to influence the thinking and action of the villagers so that they adopt better practices and improve their own life and the life of the community.

10.2 Definition of rural leadership

- **Ordway Tead:** Leadership is the activity of influencing people to cooperate towards some goal which they come to find desirable.
- **Koontz and O'Donnell:** Leadership is the art or process of influencing people so that they will strive willingly towards the achievement of group goals.
- **Terry:** Leadership is the activity of influencing people to strive willingly for group objectives.
- **Rural leadership (in extension):** Rural leadership is the process of influencing the behaviour of rural people, in a voluntary manner, to bring about desirable changes in their farm, home and community life.

The key words are: **influence, voluntary, group goal, followers, change**.

10.3 Characteristics and qualities of a good rural leader

- **Acceptance by the people:** The leader must be accepted and trusted by the villagers.
- **Vision and initiative:** The leader must see what needs to be done and take the first step.
- **Knowledge and skill:** The leader should know agriculture and village affairs well enough to guide others.
- **Communication ability:** The leader must be able to explain and persuade.
- **Service motive:** The leader should work for the good of the community, not only for personal gain.
- **Courage and patience:** The leader must face difficulties and not give up easily.
- **Cooperation:** The leader must be able to work with others and with officials.
- **Honesty and integrity:** The leader must be truthful and fair; trust is the base of leadership.
- **Impartiality:** The leader should treat all sections fairly, including the poor and the weak.

10.4 Functions of a rural leader

- To act as a **spokesman** of the village to officials and outsiders.
- To act as a **link** between the extension worker and the villagers.
- To **spread** new ideas and practices among the people.
- To **organise** meetings, groups and community activities.

- To **guide and advise** fellow farmers.
- To **represent** the village in panchayats, cooperatives and committees.
- To help in **planning and executing** extension programmes.
- To **settle disputes** and keep the community united.
- To **set an example** by adopting improved practices first (demonstration effect).

10.5 Types of leaders in the rural context

Rural leaders can be classified in several ways. The most common classifications are given below.

Basis of classification	Types	Meaning / example
Nature of authority	Formal leader	Holds an official post - sarpanch, panchayat member, cooperative chairman
Nature of authority	Informal leader	No official post, but influences by personality - an influential farmer or elder
Area of influence	Local leader	Influence confined to one village
Area of influence	Regional / state leader	Influence over a wider area or a region
Field of activity	Political leader	Active in politics and panchayat affairs
Field of activity	Religious leader	Priest or religious head who guides the community
Field of activity	Functional / specialised leader	Leads in one field - dairy, irrigation, credit, women's group
Source of emergence	Traditional leader	Leadership by inheritance, caste, age or land ownership
Source of emergence	Emergent leader	Leadership earned through ability, education or service
Attitude to change	Progressive leader	Favours change and adopts new practices
Attitude to change	Conservative leader	Opposes change and prefers old ways
Attitude to change	Neutral leader	Neither favours nor opposes change

10.5.1 Formal and informal leaders

- A **formal leader** derives authority from an official position (sarpanch, panchayat president, cooperative president, school committee head). He may or may not be truly influential.
- An **informal leader** has no official position but has real influence because people respect, trust or fear him. In extension work, informal leaders are often more useful than formal leaders because villagers follow them willingly.

10.5.2 Traditional and emergent leaders

- **Traditional leaders** owe their position to inherited factors - caste, age, family status and land ownership.
- **Emergent leaders** have earned their position through personal effort, education, ability and service to the community. Modern extension work tries to identify and develop emergent leaders.

10.5.3 Opinion leader, key communicator and change agent

- **Opinion leader:** A person whom others consult for advice on a particular topic; the first to be influenced and the first to influence others.
- **Key communicator:** A person who passes information quickly to many people; often used by extension workers.
- **Change agent:** The professional extension worker whose job is to bring about change, usually by working through local leaders.

10.6 Methods of identification of rural leaders

The extension worker must be able to identify the real leaders of a village. The important methods are given below.

1. **Sociometric method:** Villagers are asked whom they would consult, whom they would like to work with, or from whom they would take advice on a particular matter. The persons named most often are the leaders. This is the most scientific method.
2. **Key informant method:** A few knowledgeable persons (the school teacher, the VLW, the postmaster, the shopkeeper) are asked to name the leaders of the village.
3. **Observation method:** The extension worker observes the village during meetings and festivals and notes who talks, who is listened to and who takes the lead. This takes time.
4. **Reputation method:** The village is surveyed and people are asked who the "influential persons" are; those with the highest reputation are the leaders.
5. **Position / office-holding method:** Persons holding official posts (panchayat, cooperative, committees) are taken as leaders. It is easy but may miss informal leaders.
6. **Self-designating method:** Each person is asked to say how much influence he or she has; the ones who claim and are confirmed as influential are the leaders.
7. **Rural leader's nomination:** Known leaders are asked to nominate others who influence opinion in the village.
8. **Group discussion method:** In a village meeting, the extension worker notes who guides the discussion and whose opinions are accepted.

10.6.1 Comparison of the main methods

Method	How it works	Advantage	Limitation
Sociometric	Villagers name whom they consult	Most scientific and reliable	Takes time; needs skill
Key informant	A few knowledgeable persons name leaders	Quick and easy	Depends on the informant's bias
Observation	Watch the village over time	Sees real influence	Slow; needs long stay
Reputation	Survey asks who is influential	Covers the village	May favour the powerful
Position	Take official post-holders	Very easy	May miss informal leaders
Self-designating	Ask each person his influence	Simple	People may over- or under-claim

10.7 Worked example - identifying the leader from a sociometric table

Problem. In a village survey, farmers were asked whom they consult for advice on a new paddy variety. The number of times each person was named is: Ravi 12, Kumar 5, Lakshmi 9, Arul 3, Selvam 2, and the rest 0. Who is the opinion leader, and who else should the extension worker work with?

Solution. Ravi, named 12 times, is the **opinion leader** of the village for paddy. Lakshmi, named 9 times, is the **second most influential** and is important because she can reach the women of the village. The extension worker should first convince Ravi and Lakshmi and, through them, reach the other farmers - this is the principle of working through local leaders.

Exam Focus

Key points to remember

- Leadership = process of influencing people to work willingly towards a group goal; needs followers and consent.
- Rural leadership = influencing rural people voluntarily to bring desirable change.
- Qualities: acceptance, vision, knowledge, communication, service motive, courage, cooperation, honesty, impartiality.
- Functions: spokesman, link, spreader of ideas, organiser, guide, representative, planner, settler of disputes, example-setter.
- Types: formal vs informal; local vs regional; political, religious, functional; traditional vs emergent; progressive, conservative, neutral.
- Opinion leader, key communicator and change agent.
- Identification methods: sociometric, key informant, observation, reputation, position, self-designating, nomination, group discussion.
- Sociometric method is the most scientific; position method is easiest but may miss informal leaders.

Short-answer questions with model answers

Q1. Define leadership and rural leadership.

Ans. Leadership is the process of influencing people to cooperate willingly towards a group goal (Tead; Koontz and O'Donnell). Rural leadership is the process of influencing rural people voluntarily to bring about desirable changes in their farm, home and community life.

Q2. Differentiate formal and informal leaders.

Ans. A formal leader holds an official post (sarpanch, cooperative president) and has authority from the position; an informal leader has no official post but real influence through personality, trust or respect. Informal leaders are often more useful in extension.

Q3. Name any four methods of identifying rural leaders.

Ans. Sociometric method, key informant method, observation method and reputation method; also position, self-designating, nomination and group discussion methods.

Q4. What is an opinion leader?

Ans. An opinion leader is a person whom others consult for advice on a particular topic, who is among the first to be influenced and the first to influence others in the group.

Long-answer questions with model answer outlines

Q1. Explain the concept and definition of rural leadership, the types of leaders and the methods of identifying them.

Outline. Define leadership and rural leadership. List qualities and functions. Explain types with the classification table - formal/informal, local/regional, political/religious/functional, traditional/emergent, progressive/conservative/neutral. Describe the methods of identification with the comparison table and state that the sociometric method is the most scientific. Give a sociometric worked example. Conclude with the principle of working through local leaders.

Q2. Discuss the role of rural leaders in extension work and how an extension worker can identify them.

Outline. Explain why leaders matter - trust, low cost, reach, demonstration effect. List the functions of a rural leader. Describe identification methods and their merits and limits. Give an example of using an opinion leader to spread a new practice. Conclude with the need to develop emergent leaders, including women.

Chapter 11. Extension Administration; Monitoring and Evaluation

11.1 Extension administration: meaning and concept

Administration means the process of planning, organising, directing, coordinating and controlling the resources of an organisation to achieve its objectives. **Extension administration** is the administration of an extension organisation - it is the process of getting the extension work done efficiently through people, by planning, organising, staffing, directing, coordinating and controlling the extension activities.

In simple terms, extension administration is the **management of the extension system** so that the extension programme reaches the farmer effectively. It deals with the "how" of running the extension machinery: how staff are posted, how work is planned and supervised, how funds and materials are used, and how the results are checked.

11.2 Principles of extension administration

- **Principle of unity of command:** Each worker should receive orders from one superior only, to avoid confusion.
- **Principle of delegation:** Authority and responsibility should be delegated to lower levels so that decisions can be taken quickly.
- **Principle of span of control:** A superior should supervise only a limited number of subordinates (a manageable span).
- **Principle of coordination:** All departments and workers should work together towards common objectives.
- **Principle of efficiency:** The organisation should achieve its objectives with the least cost and effort.
- **Principle of flexibility:** The administrative set-up must be able to change with changing conditions.
- **Principle of democratic leadership:** Administration should be democratic, participative and human.
- **Principle of effective communication:** Orders and information must flow clearly both downward and upward.
- **Principle of the line and staff:** There should be a clear line of authority, supported by staff (specialists) who advise.
- **Principle of centralisation of policy and decentralisation of execution:** Policy is decided at the top, but execution is done at the field level.

11.3 Functions of extension administration

The functions of extension administration follow the classical management functions - **POSDCORB**, a useful memory aid.

- **P - Planning:** Deciding in advance what to do, how, when and by whom.
- **O - Organising:** Setting up the structure, defining duties and arranging resources.
- **S - Staffing:** Recruiting, posting, training and managing personnel.

- **D - Directing:** Leading, guiding, motivating and supervising the staff.
- **CO - Coordinating:** Bringing together the work of different units and agencies.
- **R - Reporting:** Keeping records and reporting progress to higher levels.
- **B - Budgeting:** Preparing and controlling the budget and funds.

In addition, extension administration performs the following special functions:

- Maintaining **research-extension-farmer linkage**.
- Ensuring the **timely supply** of inputs and materials.
- Arranging **training** for extension staff.
- Maintaining **records, reports and statistics**.
- **Supervising** field work through visits and inspections.

11.4 Levels of extension administration in India

Level	Unit	Main officer (indicative)
State	State Department of Agriculture	Director of Agriculture
District	District	Joint / Deputy Director of Agriculture
Block	Block	Assistant Director / Agricultural Officer
Village	Village	Assistant Agricultural Officer / Village Extension Worker

The same pattern works in TNAU's extension wing, with the Directorate of Extension Education at the top, KVKs at the district level, and subject-matter specialists linking research to the field.

11.5 Monitoring and evaluation: concept and definition

Monitoring is the continuous and systematic collection of information on the progress of an ongoing programme, to check whether the work is going as planned and to take corrective action in time. It is like watching the dials while the machine is running.

Evaluation is the periodic and systematic assessment of the worth, value or results of a programme, to judge whether the objectives were achieved and how much. It is like checking the final product and asking whether it was worth the effort.

Definitions:

- **Monitoring:** Monitoring is the continuous supervision and checking of the progress of a programme to ensure that it is on the right track (also called process or progress evaluation).
- **Evaluation:** Evaluation is the process of determining the extent to which the objectives of a programme have been achieved (Tyler); it is the systematic assessment of the value of a programme.

11.5.1 Difference between monitoring and evaluation

Point	Monitoring	Evaluation
Nature	Continuous, ongoing	Periodic, at intervals or at the end
Purpose	Check progress; take timely corrective action	Judge results and worth; take decisions

Point	Monitoring	Evaluation
Timing	During implementation	Mid-term, at the end, or after
Focus	Inputs, activities, outputs	Outcomes, impact, achievement of objectives
Question	"Are we doing things right?"	"Are we doing the right things, and did they work?"
Who does it	Programme staff / supervisors	Independent or higher-level evaluators
Example	Monthly report of demonstrations held	Study of yield change after one season

11.6 Monitoring and evaluation of extension programmes

Monitoring and evaluation of extension programmes involves the following steps:

1 **Fix the objectives and the indicators:** Decide what is to be measured (for example, number of demonstrations, number of adopters, change in yield).

2 **Set a baseline:** Record the situation before the programme (the "before" picture).

3 **Collect data regularly:** Through reports, records, visits, interviews and observations.

4 **Compare with the plan:** Compare actual progress with the planned targets.

5 **Analyse the results:** Find out what worked, what did not, and why.

6 **Take corrective action:** Modify the programme during monitoring; draw lessons during evaluation.

7 **Report and use findings:** Feed the findings back into planning for the next programme.

11.6.1 Criteria and indicators of evaluation

- **Relevance:** Was the programme suited to the real needs of the people?
- **Effectiveness:** Were the objectives achieved?
- **Efficiency:** Was the output obtained at reasonable cost?
- **Impact:** What lasting change occurred in the lives of the people?
- **Sustainability:** Will the benefits continue after the programme ends?

11.7 Types of evaluation

Evaluation can be classified in several ways.

Basis	Type	Meaning
Time of conduct	Ex-ante (pre-evaluation)	Before the programme starts; appraises the plan
Time of conduct	Concurrent (on-going)	During the programme; same as monitoring
Time of conduct	Ex-post (post-evaluation)	After the programme; judges final results
Purpose	Formative	To improve the programme while it is running
Purpose	Summative	To judge the overall success at the end

Basis	Type	Meaning
Who evaluates	Internal	By the programme's own staff
Who evaluates	External	By outside or independent evaluators
Scope	Process evaluation	Studies how the programme was carried out
Scope	Impact evaluation	Studies the final change or benefit
Basis of comparison	Before-and-after	Compares the situation before and after
Basis of comparison	With-and-without	Compares participants with non-participants

11.7.1 Worked example - a before-and-after evaluation

Problem. In a village, 40 farmers grew paddy. Before a KVK demonstration programme the average yield was 4.0 tonnes per hectare. After the programme, 28 of the 40 farmers adopted the recommended practice, and the average yield of the adopters rose to 5.2 tonnes per hectare. Evaluate the outcome.

Solution.

- Adoption percentage = (28 / 40) x 100 = **70 per cent**.
- Increase in yield of adopters = 5.2 - 4.0 = **1.2 tonnes per hectare**.
- Percentage increase = (1.2 / 4.0) x 100 = **30 per cent**.
- The programme is effective: adoption is high (70%) and the yield gain (30%) is substantial. Efficiency and sustainability would need cost data and a follow-up after the next season.

11.8 Importance of monitoring and evaluation

- They show whether the programme is on track and whether objectives are being achieved.
- They provide the basis for taking timely corrective action.
- They give feedback to planners, researchers and extension workers.
- They justify the use of funds and build accountability.
- They provide evidence of success, which helps to secure support and to spread good practices.
- They help to learn from experience and to improve future programmes.

Exam Focus

Key points to remember

- Extension administration = management of the extension organisation to achieve its objectives through people.
- POSDCORB: Planning, Organising, Staffing, Directing, Coordinating, Reporting, Budgeting.
- Principles: unity of command, delegation, span of control, coordination, efficiency, flexibility, democratic leadership, communication, line and staff, centralised policy and decentralised execution.
- Monitoring = continuous checking of progress during a programme; evaluation = periodic judgement of results.

- Monitoring asks "are we doing things right?"; evaluation asks "did the programme work?".
- Types of evaluation: ex-ante, concurrent, ex-post; formative, summative; internal, external; process, impact; before-after, with-without.
- Criteria: relevance, effectiveness, efficiency, impact, sustainability.

Short-answer questions with model answers

Q1. Define extension administration.

Ans. Extension administration is the process of planning, organising, directing, coordinating and controlling the resources of an extension organisation so that the extension programme reaches the farmer effectively and its objectives are achieved.

Q2. What is POSDCORB?

Ans. A memory aid for the functions of administration: Planning, Organising, Staffing, Directing, Coordinating, Reporting and Budgeting.

Q3. Differentiate monitoring from evaluation.

Ans. Monitoring is continuous checking of progress during a programme to take timely corrective action; evaluation is periodic assessment of results and worth to judge whether objectives were achieved.

Q4. Name any four types of evaluation.

Ans. Ex-ante, concurrent, ex-post; formative and summative; internal and external; process and impact; before-and-after and with-and-without.

Long-answer questions with model answer outlines

Q1. Explain the meaning, principles and functions of extension administration.

Outline. Define administration and extension administration. Explain the principles (unity of command, delegation, span of control, coordination, efficiency, flexibility, democratic leadership, communication, line and staff). Explain the functions using POSDCORB, plus special functions such as research linkage, input supply and training. Show the levels of administration in India and Tamil Nadu. Conclude with the role of good administration in effective extension.

Q2. Define monitoring and evaluation. Explain the types of evaluation and the steps in evaluating an extension programme.

Outline. Define monitoring and evaluation; give the comparison table. Explain the steps - fix objectives and indicators, set baseline, collect data, compare, analyse, correct, report. List the criteria (relevance, effectiveness, efficiency, impact, sustainability). Explain the types of evaluation with the table. Solve a before-and-after example. Conclude with the importance of monitoring and evaluation.

Chapter 12. Transfer of Technology; Capacity Building

12.1 Concept of transfer of technology

Transfer of technology (TOT) is the process by which the results of research - new knowledge, techniques, inputs and practices - are carried from the research system to the farmer and are adopted by the farmer in his field. It is the bridge between the laboratory and the land.

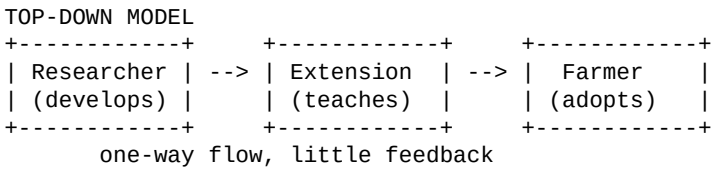
TOT is not a single act of handing over information; it is a **continuous, two-way process** in which technology flows to the farmer and the farmer's problems and feedback flow back to the researcher. The success of TOT is measured not by how much information was sent out, but by how much technology was actually **adopted** and how much benefit the farmer got.

12.2 Models of transfer of technology

Several models describe how technology moves from research to the farmer. The four important models are the Top-down Model, the Feedback Model, the Farmer-back-to-Farmer Model and the REP (Research-Extension-Producer) Interaction Model.

12.2.1 Top-down model (transfer-of-technology model)

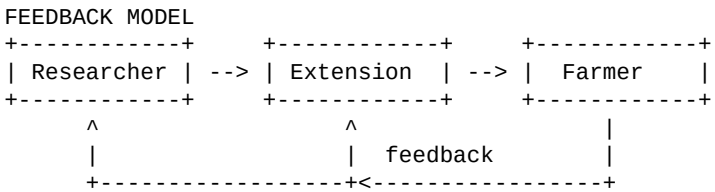
In the top-down model, the researcher develops the technology, the extension worker passes it to the farmer, and the farmer is expected to adopt it. The flow is one-way: research to extension to farmer. The farmer is a passive receiver.



Merits: Simple, quick, and useful when the technology is clearly superior and the farmer's situation is uniform. **Demerits:** The farmer's problems and local conditions are ignored; technology may not fit the farmer's needs; the farmer is treated as a passive receiver; adoption may fail.

12.2.2 Feedback model

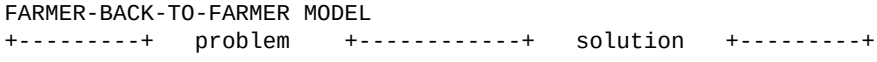
The feedback model is a correction of the top-down model. It keeps the flow from research to farmer but adds a **feedback channel** from the farmer back to the extension worker and researcher, so that the technology can be modified.

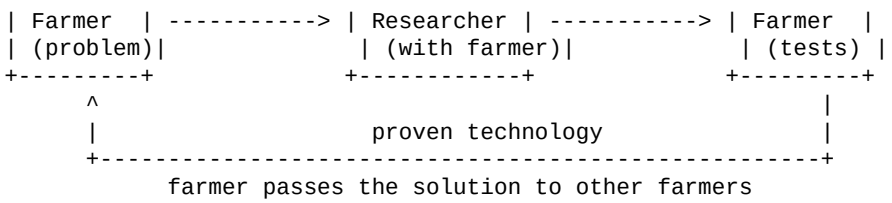


Merits: Two-way; the researcher learns the farmer's problems and adjusts the technology. **Demerits:** The feedback is often weak or slow; the basic direction is still from top to bottom.

12.2.3 Farmer-back-to-farmer model

In the farmer-back-to-farmer model, the **farmer is the starting point**. The researcher first studies the farmer's real problem, practices and constraints, then develops a solution **with** the farmer, tests it in the farmer's field, and finally the farmer passes the proven solution to other farmers. The farmer is an active partner in research and extension.

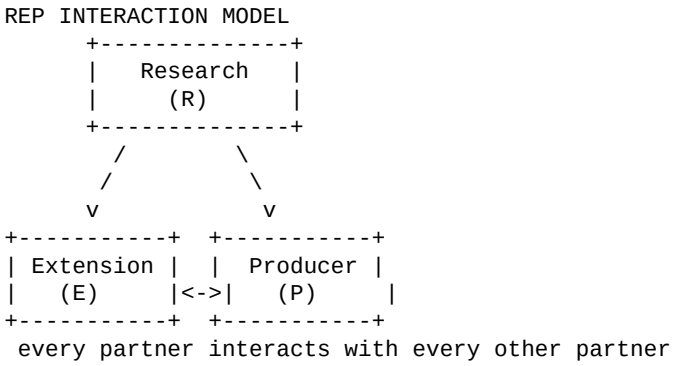




Merits: Technology fits the farmer's real needs; the farmer participates; adoption is high. **Demerits:** Slow and demanding; needs close research-farmer contact; not suitable for quick, mass transfer.

12.2.4 REP interaction model (Research-Extension-Producer)

The REP model shows the interaction among three partners - **Research (R), Extension (E) and Producer/farmer (P)**. Each partner interacts with the other two, and the technology moves through their interaction. It recognises that research, extension and the farmer are equal partners in a system, not a one-way chain.



Merits: Recognises the farmer as a partner; promotes two-way flow among all three; suits participatory and market-led extension. **Demerits:** Requires good coordination and communication among the three partners.

12.2.5 Comparison of the four TOT models

Model	Direction of flow	Role of farmer	Main strength	Main weakness
Top-down	Research to extension to farmer (one-way)	Passive receiver	Simple and quick	Ignores farmer's needs
Feedback	Top-down with a feedback loop	Receiver who gives feedback	Corrects technology through feedback	Feedback weak and slow
Farmer-back-to-farmer	Farmer to researcher to farmer	Active partner	Fits real needs; high adoption	Slow and demanding
REP interaction	Among R, E and P (two-way)	Equal partner	Balanced two-way system	Needs strong coordination

12.3 Capacity building of extension personnel

Capacity building means the process of developing the knowledge, skills, attitudes and abilities of extension personnel so that they can perform their work effectively and keep pace with new technology. Capacity building is done mainly through **training**, along with exposure visits, workshops and study tours.

12.4 Training: meaning and definition

Training is a planned and systematic process of developing the knowledge, skills and attitudes of a person so that he or she can perform a particular job or task more effectively. Training is job-oriented; it is aimed at improving performance in a specific role.

Definitions:

- **Flippo:** Training is the act of increasing the knowledge and skill of an employee for doing a particular job.
- **Beach:** Training is the organised procedure by which people learn knowledge and skill for a definite purpose.
- **KVK idea of training:** Training is the process of bringing about a change in the behaviour of the learner in a practical, "learning by doing" manner.

12.4.1 Difference between training and education

Point	Training	Education
Aim	Improve performance in a specific job	Develop the whole personality
Scope	Narrow, job-specific	Broad, general
Duration	Short, repeated	Long, continuous
Focus	Skills and practical ability	Knowledge, skills and attitudes together
Result	Better job performance	Better citizen and all-round development
Example	KVK training on drip irrigation	B.Sc. Agriculture degree programme

12.5 Types of training

Training may be classified in several ways.

Basis	Type	Meaning / example
Who is trained	Pre-service training	Before joining the job - induction and foundation courses
Who is trained	In-service training	While in service, to update knowledge - refresher courses
Who is trained	Orientation training	To familiarise a new worker with the organisation and job
Who is trained	Induction training	Given to a new recruit on entry
Duration	Short-duration	A few days to a few weeks
Duration	Long-duration	Several weeks to months
Content	Skill training	To develop a manual or technical skill
Content	Knowledge / subject training	To update subject knowledge
Content	Attitudinal training	To change attitudes and motivation
Method	On-the-job	Learning while doing the actual job, under supervision
Method	Off-the-job	Learning away from the job - classroom, laboratory, field

Basis	Type	Meaning / example
Purpose	Refresher training	To refresh and update existing knowledge
Purpose	Promotional training	To prepare a person for a higher post

12.5.1 Methods of training

- **Lecture and discussion:** For knowledge and theory.
- **Demonstration:** For skills - showing then getting the learner to do.
- **Practical and laboratory work:** Hands-on learning.
- **Field visits and study tours:** Learning from other farms and villages.
- **Case study and role play:** For attitudes and decision-making.
- **Simulation and games:** For problem-solving.
- **Programmed instruction and e-learning:** Self-paced learning.

12.6 Worked example - designing a training programme for a KVK

Problem. A KVK wants to train 30 farm women of a village in preparing value-added products from banana (chips, jam, halwa) so that they can earn extra income. Suggest a simple training design.

Solution.

Element	Design
Trainees	30 farm women of the village
Objective	To enable each trainee to prepare and pack three banana products and calculate their cost
Duration	Three days (short-duration, skill training)
Content	Selection and cleaning, recipes, hygienic handling, packing, costing, marketing
Methods	Demonstration, hands-on practice, group discussion, field visit to a unit
Materials	Banana, sugar, oil, packing material, stove, weighing scale
Trainers	Home science specialist, food technology specialist, a successful entrepreneur
Evaluation	Each trainee prepares the products and is assessed on a checklist; follow-up visit after one month

Exam Focus

Key points to remember

- TOT = carrying research results to the farmer and getting them adopted; a continuous two-way process.
- Four models: top-down (one-way), feedback (with a loop), farmer-back-to-farmer (farmer as starting point), REP interaction (R-E-P partners).
- Top-down is simple but ignores the farmer; farmer-back-to-farmer gives high adoption but is slow.
- Capacity building = developing knowledge, skills and attitudes of extension personnel, mainly through training.
- Training = planned, job-oriented process of increasing knowledge and skill (Flippo; Beach).

- Training differs from education: job-specific and short vs broad and long.
- Types: pre-service and in-service; orientation, induction, refresher, promotional; short and long; on-the-job and off-the-job; skill, knowledge and attitudinal.

Short-answer questions with model answers

Q1. Define transfer of technology.

Ans. Transfer of technology is the process by which research results are carried to the farmer and adopted in the field; it is a continuous two-way process in which technology goes to the farmer and the farmer's problems go back to the researcher.

Q2. Name the four models of transfer of technology.

Ans. Top-down model, feedback model, farmer-back-to-farmer model and REP (Research-Extension-Producer) interaction model.

Q3. Define training and give one definition.

Ans. Training is a planned and systematic process of developing the knowledge, skills and attitudes of a person for a particular job. Flippo: "Training is the act of increasing the knowledge and skill of an employee for doing a particular job."

Q4. Differentiate pre-service and in-service training.

Ans. Pre-service training is given before joining the job (foundation and induction courses); in-service training is given while in service to update knowledge and skills (refresher courses).

Long-answer questions with model answer outlines

Q1. Explain the concept of transfer of technology and describe its important models.

Outline. Define TOT and state that it is two-way. Explain each model with its diagram - top-down, feedback, farmer-back-to-farmer and REP interaction. Present the comparison table. Give the merits and demerits of each. Conclude that modern extension favours participatory models such as farmer-back-to-farmer and REP.

Q2. What is capacity building? Explain the meaning, definition and types of training for extension personnel.

Outline. Define capacity building. Define training with two definitions. Distinguish training from education. Explain the types of training with the table - pre-service and in-service, orientation and induction, refresher and promotional, short and long, on-the-job and off-the-job, skill, knowledge and attitudinal. Describe training methods. Give a KVK training design example. Conclude with the importance of capacity building for effective extension.

Chapter 13. Extension Teaching Methods - Individual Contact Methods

13.1 Meaning of extension teaching methods

An extension teaching method is the **device or technique** by which the extension worker brings the farmer and the recommended practice together so that the farmer learns and adopts it. A method is the "how" of teaching - it is the means by which the teaching-learning situation is created.

Teaching in extension is not a single act. It involves three essential elements, often called the **teaching-learning situation**:

- A **teacher** - the extension worker or a local leader.
- A **learner** - the farmer, farm woman or youth.
- A **subject matter** - the recommended practice or information.
- A **teaching method** - the device that brings all three together.

13.2 Classification of extension teaching methods

Extension teaching methods are classified according to the **number of people reached** and the **nature of contact**. The three main classes are:

1. **Individual contact methods** - one person at a time (farm and home visit, personal letter, official call, telephone call, result demonstration).
2. **Group contact methods** - a small group at a time (method demonstration, meeting, lecture, discussion, etc.).
3. **Mass contact methods** - many people at a time (campaign, exhibition, radio, television, print media).

Class	People reached	Contact	Examples	Main use
Individual	One at a time	Face-to-face or personal	Farm and home visit, personal letter, official call	To advise, convince and get adoption from specific farmers
Group	A few to a few dozen	Face-to-face with a group	Method demonstration, meeting, lecture	To teach skills and knowledge to a group
Mass	Many at a time	Impersonal	Campaign, exhibition, radio, TV, print	To create awareness and spread information quickly

13.2.1 How to select a method

The choice of method depends on:

- The **objective** - awareness, knowledge, skill or attitude change.
- The **number of people** to be reached.
- The **nature of the practice** - simple or complex, single or a package.

- The **time and resources** available.
- The **level of the people** - literacy, education, interest.
- The **stage of adoption** - awareness, interest, evaluation, trial or adoption.

13.3 Individual contact methods

Individual contact methods involve face-to-face or personal communication with a single farmer or family. They are the most effective methods for **convincing** a farmer and for getting a **decision to adopt**, because the teaching can be fitted to the individual's situation. Their weakness is that they are **time-consuming** and reach only a few people.

13.3.1 Farm and home visit

A farm and home visit is a visit by the extension worker to the farmer's **farm and home** to advise him on his problems. The farm visit deals with crop, livestock and field problems; the home visit deals with the family, home, health and nutrition, and is the best way to reach farm women.

Purpose: To give personal advice, to understand the farmer's real problems and resources, to build confidence and friendship, to follow up a previous programme, and to persuade the farmer to adopt a practice.

Principles of a good visit:

- Visit at a time convenient to the farmer, not the worker.
- Have a clear purpose and plan the visit in advance.
- Be friendly, respectful and informal; be a guest, not an inspector.
- Listen more than you talk; start from the farmer's problem.
- Do not criticise the farmer or his practices.
- Give advice that fits the farmer's resources.
- Keep the visit short and useful; leave a clear suggestion.
- Follow up later to check progress.

Advantages: Personal and effective; builds confidence and friendship; advice can be fitted to the individual; good for follow-up and for reaching women.

Limitations: Very time-consuming; reaches only one person at a time; costly per contact; the worker may not be able to visit every farmer.

13.3.2 Personal letter

A personal letter is a letter written by the extension worker to a particular farmer or family, giving advice, information or an invitation. It may be typed or, better, handwritten for a personal touch.

Advantages: Gives a personal touch; reaches people who cannot be visited; a written record that can be read again; cheap; useful for a farmer who is literate and away from the village.

Limitations: Useful only for literate farmers; one-way; slow; cannot answer the farmer's questions immediately; may not be read.

13.3.3 Official call (office call)

An official call is a visit by the farmer to the **office of the extension worker** to ask for advice or help, or a call by the extension worker to the farmer's house on official business. When the farmer comes to the office, the extension worker must receive him courteously, listen to his problem and give clear advice.

Advantages: The farmer comes with a real, felt problem, so he is ready to learn; no travel for the worker; useful for urgent problems; the office becomes a place of help.

Limitations: Only the farmers who come are helped; the poor and shy may not come; the worker cannot choose the topic.

13.3.4 Observation (and telephone call)

Observation means the extension worker carefully observing the farmer's fields, livestock, home and practices to understand his problems and to plan advice. It is used along with the farm and home visit.

A **telephone call** (or mobile call) is a personal contact at a distance. It is quick, useful for urgent advice, and saves travel, but it depends on the telephone and on the farmer being reachable.

13.3.5 Result demonstration

A result demonstration is a demonstration that shows the **results** of a practice - the difference between the improved practice and the old practice - so that the farmer can see the advantage for himself. It is "**learning by seeing**". For example, two plots of paddy are grown side by side, one with the recommended practice and one with the farmer's practice, and the yields are compared at harvest.

Purpose: To convince the farmer by showing visible proof; to show the value of a practice; to remove doubts and fears; to create confidence in the extension worker and the technology.

Steps in conducting a result demonstration:

1 Select a **cooperative farmer** with a suitable field and a genuine problem.

2 **Plan** the demonstration - what practice, what plot, what records.

3 Arrange the **inputs and materials**.

4 Lay out a **demonstration plot and a control (check) plot** side by side.

5 **Record** the inputs, operations and yields of both plots.

6 **Conduct** the demonstration with the farmer's own hands where possible.

7 **Hold a field day** at harvest to show the results to other farmers.

8 **Evaluate** and report the results.

Advantages: Very convincing - seeing is believing; uses the farmer's own field; creates confidence; useful for showing the value of a whole package; the farmer's own labour makes it his demonstration.

Limitations: Takes a whole season; needs careful planning and record-keeping; results can be spoilt by weather or pests; costly; reaches a limited number of people.

13.3.6 Method demonstration and result demonstration compared

Point	Method demonstration	Result demonstration
Shows	"How to do" a practice	"What happens" if the practice is used
Duration	Short - minutes to an hour	Long - a whole season or crop cycle
Main purpose	Teach a skill	Convince by showing results
Place	Any convenient place	In the field, with a control plot
Learning	Learning by doing	Learning by seeing
Example	Showing how to treat seed	Comparing yield of treated and untreated plots

13.4 Worked example - cost-benefit of a result demonstration

Problem. A result demonstration on paddy was laid out on 0.1 hectare each for the demonstration plot (recommended practice) and the control plot (farmer's practice). The recommended practice cost Rs. 3,000 extra in inputs but gave a yield of 700 kg, against 500 kg in the control plot. Paddy sells at Rs. 22 per kg. Work out the additional benefit and the benefit-cost ratio of the demonstration.

Solution.

Item	Control (farmer's practice)	Demonstration (recommended)
Yield	500 kg	700 kg
Value of produce (Rs. 22/kg)	500 x 22 = Rs. 11,000	700 x 22 = Rs. 15,400
Additional input cost	-	Rs. 3,000
Net value	Rs. 11,000	15,400 - 3,000 = Rs. 12,400

- Additional yield = 700 - 500 = **200 kg**.
- Additional gross return = 200 x 22 = **Rs. 4,400**.
- Additional net benefit = 4,400 - 3,000 = **Rs. 1,400**.
- Benefit-cost ratio = additional gross return / additional cost = 4,400 / 3,000 = **1.47**.

Conclusion: For every one rupee of extra input cost, the farmer gains about Rs. 1.47 of extra produce value. The demonstration is profitable and can be shown to other farmers.

Exam Focus

Key points to remember

- Extension teaching method = device by which the worker brings the farmer and the practice together.
- Teaching-learning situation: teacher, learner, subject matter and method.
- Classification by number reached: individual, group and mass contact methods.
- Individual methods: farm and home visit, personal letter, official call, telephone call, observation, result demonstration.
- Farm and home visit is the best method for convincing a farmer and for reaching women.
- Result demonstration = learning by seeing; shows the difference between improved and old practice; needs a control plot and a field day.

- Method demonstration shows "how", result demonstration shows "what happens".
- Individual methods are effective but time-consuming and costly per contact.

Short-answer questions with model answers

Q1. What is an extension teaching method?

Ans. An extension teaching method is the device or technique by which the extension worker brings the farmer and the recommended practice together so that the farmer learns and adopts it.

Q2. Classify extension teaching methods.

Ans. By the number of people reached: individual contact methods (one at a time), group contact methods (a group at a time) and mass contact methods (many at a time).

Q3. What is a result demonstration? Give one example.

Ans. A result demonstration shows the results of a practice by comparing an improved plot with a control plot, so that the farmer sees the advantage for himself. Example: comparing yield of paddy grown with the recommended package against the farmer's practice.

Q4. State any four principles of a good farm and home visit.

Ans. Visit at the farmer's convenience, have a clear purpose, be friendly and respectful, listen to the farmer's problem first; also give advice that fits his resources and follow up later.

Long-answer questions with model answer outlines

Q1. Explain the meaning and classification of extension teaching methods and describe the individual contact methods.

Outline. Define teaching method and the teaching-learning situation. Explain the classification (individual, group, mass) with the table. Describe each individual method - farm and home visit, personal letter, official call, telephone call, observation and result demonstration - with meaning, purpose, advantages and limitations. Give the steps of a result demonstration. Present the method vs result demonstration table and the cost-benefit example. Conclude that individual methods are effective but limited in reach.

Q2. What is a result demonstration? Explain its steps, advantages and limitations, and distinguish it from a method demonstration.

Outline. Define result demonstration as learning by seeing. Explain the purpose and the eight steps. List advantages and limitations. Present the comparison table with method demonstration. Solve the cost-benefit worked example. Conclude that it is the most convincing method for a new practice.

Chapter 14. Extension Teaching Methods - Group Contact Methods

14.1 Meaning of group contact methods

Group contact methods are teaching methods in which the extension worker teaches a **group of people** - usually a small number, from about five to thirty - at the same time, in the same place. They are more efficient than individual methods because they reach more people per unit of the worker's time, and they also allow the group members to learn from one another.

Group methods are especially good for teaching **skills** (method demonstration), for **discussion and decision-making** (meeting, workshop, forum, panel) and for **giving information** (lecture, symposium, conference). Their weakness is that they give less personal attention than individual methods.

14.2 Important group contact methods

14.2.1 Method demonstration

A method demonstration shows the group **how to do** a particular job, step by step - for example, how to treat seed, how to prune, how to prepare a nursery, how to use a sprayer. It is "**learning by doing**". The worker shows the steps and then asks members of the group to repeat them.

Advantages: Teaches a skill effectively; the learner can see and do; the worker can correct mistakes at once; good for a group; less costly than individual teaching. **Limitations:** Needs materials and preparation; the group may not all see clearly; results are not shown, only the method.

14.2.2 Meeting (group meeting)

A meeting is a gathering of people for a definite purpose - to discuss a problem, to plan an activity or to decide something. In extension, meetings are used for planning, for explaining a programme and for taking group decisions.

Advantages: Allows discussion and participation; reaches many people; helps in planning and decision-making; builds group feeling. **Limitations:** Needs good leadership and organisation; a few may dominate; decisions may be vague.

14.2.3 Lecture

A lecture is a **formal talk** by the extension worker to a group, with little or no participation by the audience. It is one-way communication, good for giving information quickly to many people.

Advantages: Reaches many people; quick; useful for giving facts and information; no special equipment needed. **Limitations:** One-way; the audience is passive; poor for teaching skills and for changing attitudes; the attention of the audience may wander.

14.2.4 Debate

A debate is a **formal discussion** in which two sides argue for and against a proposition, and one side wins by argument. In extension, a debate can be used to bring out the pros and cons of a new practice and to interest people in a subject.

Advantages: Develops thinking and speaking; brings out both sides of a question; interesting and lively. **Limitations:** May create heat rather than light; one side may win though wrong; needs a good moderator.

14.2.5 Workshop

A workshop is a **practical training session** in which a group works together, usually in small groups, on a real problem or task and produces an output (a plan, a report, a product). It combines learning with doing.

Advantages: Practical and participatory; produces a concrete output; develops skills and teamwork. **Limitations:** Needs time, resources and good facilitation; may be dominated by the active few.

14.2.6 Seminar

A seminar is a meeting in which one or a few persons **present a paper or a topic** and the rest discuss it. It is used in extension training to develop the ability to study, present and discuss a subject.

Advantages: Develops study, presentation and discussion skills; allows deep treatment of a topic. **Limitations:** Needs prepared speakers; may be too academic for village use.

14.2.7 Forum

A forum is a meeting at which a subject is discussed **freely by all the members**, with a chairman to guide the discussion. Unlike a lecture, everyone can speak; unlike a debate, the aim is to understand, not to win.

Advantages: Free and full discussion; develops participation and thinking; useful for solving problems. **Limitations:** Needs a good chairman; may become disorderly; a few may talk too much.

14.2.8 Conference

A conference is a **large meeting of people with a common interest** (often representatives from different places) to discuss a subject and take decisions or make recommendations. It is bigger than a seminar and usually covers several topics.

Advantages: Brings together many people and views; useful for policy and coordination; decisions have wide acceptance. **Limitations:** Large and formal; little individual participation; costly.

14.2.9 Symposium

A symposium is a meeting in which **several experts** speak one after another on different aspects of a single subject, and then the audience may ask questions. Each speaker presents a different point of view or a different part of the topic.

Advantages: Brings expert and balanced views on one subject; useful for a complex topic; saves time. **Limitations:** The audience is mostly passive; no free discussion among speakers; may be too technical.

14.2.10 Panel

A panel is a meeting in which a **small group of experts (the panel)** sits before an audience and discusses a subject among themselves, guided by a chairman; then the audience may put questions. It is a "public conversation" by experts.

Advantages: Informal expert discussion; different views are brought out; the audience can ask questions. **Limitations:** The panel may not agree; the audience is mostly passive; needs a skilled chairman.

14.2.11 Brainstorming

Brainstorming is a technique in which a group is asked to **produce as many ideas as possible** on a problem, freely and without criticism; the ideas are recorded and evaluated later. The aim is to generate many ideas, including new and unusual ones.

Advantages: Produces many ideas; encourages creativity; everyone can take part; no idea is criticised at the time. **Limitations:** Needs a skilled leader; the ideas must be sorted and tested later; may produce many useless ideas.

14.2.12 Buzz session (buzz group)

A buzz session is a technique in which a large group is **divided into small groups of five or six**, each discussing a question for a few minutes and reporting its conclusions to the whole group. The "buzz" of many small groups talking gives it the name.

Advantages: Gives everyone a chance to speak; quick and lively; produces many viewpoints; good for a large meeting. **Limitations:** Needs organisation and time control; the reports may repeat; needs a good summariser.

14.2.13 Role playing

Role playing is a technique in which members of the group **act out a real-life situation** (for example, a farmer talking to a bank officer), each taking a role, and then the group discusses what happened. It develops understanding of other people's views and of social situations.

Advantages: Develops understanding, empathy and communication; interesting and lively; good for attitudes and human relations. **Limitations:** Some people are shy; may become artificial; needs a skilled leader to draw lessons.

14.2.14 Simulation games

A simulation game is a technique in which the group plays a **game that imitates a real situation** (for example, a farm-management game in which players decide crops and inputs and see the "results"), so that they learn decision-making in a safe way.

Advantages: Teaches decision-making and problem-solving; interesting and involving; safe way to learn from mistakes. **Limitations:** Needs preparation and materials; the game may be too simple for reality; time-consuming.

14.3 Comparison of important group methods

Method	Nature	Participation	Best used for
Method demonstration	Showing a skill	Active - doing	Teaching a skill
Lecture	Formal talk	Passive	Giving information quickly
Meeting	Discussion for a purpose	Active	Planning and decisions
Seminar	Paper presentation + discussion	Moderate	Study and discussion
Forum	Free discussion	Very active	Understanding a subject
Symposium	Experts speak on aspects	Passive	Balanced expert views
Panel	Experts discuss before audience	Mostly passive	Informal expert views
Brainstorming	Free generation of ideas	Very active	Producing new ideas
Buzz session	Small groups in a large meeting	Very active	Involving everyone
Role playing	Acting out a situation	Active	Attitudes and human relations
Simulation game	Imitating a real situation	Active	Decision-making

Memory tip: For group methods, remember the "P-A-T" idea - **P**assive methods (lecture, symposium, panel, conference) give information; **A**ctive methods (forum, meeting, brainstorming, buzz, role play) develop participation; **T**raining methods (method demonstration, workshop, simulation) develop skills and decisions.

Exam Focus

Key points to remember

- Group contact methods teach a group at one time; more efficient than individual methods; less personal attention.
- Method demonstration shows "how" (learning by doing); best for skills.
- Lecture is one-way; forum is free two-way discussion; symposium has experts on aspects; panel has experts in discussion.
- Brainstorming generates many ideas without criticism; buzz session divides a large group into small ones.
- Role playing acts out a real situation; simulation games imitate a real situation for decision-making.
- Active methods develop participation; passive methods give information; training methods develop skills.

Short-answer questions with model answers

Q1. What are group contact methods?

Ans. Group contact methods are teaching methods in which the extension worker teaches a group of people at the same time in the same place, reaching more people than individual methods and allowing members to learn from one another.

Q2. Distinguish between a forum and a lecture.

Ans. A lecture is a formal one-way talk with a passive audience; a forum is a free discussion in which all members can speak, guided by a chairman, with the aim of understanding a subject.

Q3. What is a buzz session?

Ans. A buzz session is a technique in which a large group is divided into small groups of five or six, each discussing a question for a few minutes and then reporting its conclusions to the whole group.

Q4. What is the difference between role playing and a simulation game?

Ans. In role playing, members act out a real-life situation, each taking a role, to develop understanding of human relations; in a simulation game, the group plays a game that imitates a real situation to learn decision-making safely.

Long-answer questions with model answer outlines

Q1. Explain the important group contact methods used in extension education.

Outline. Define group contact methods and state their merits and limits. Describe each method - method demonstration, meeting, lecture, debate, workshop, seminar, forum, conference, symposium, panel, brainstorming, buzz session, role playing and simulation games - with meaning, advantages and limitations. Present the comparison table. Conclude with the P-A-T idea for choosing a method.

Q2. Distinguish between individual, group and mass contact methods, and explain any four group methods in detail.

Outline. Give the classification table (people reached, contact, examples, main use). Then explain four group methods in detail, e.g. method demonstration, forum, brainstorming and role playing, with advantages and limitations. Conclude with the principle of using a combination of methods according to the objective.

Chapter 15. Mass Contact Methods; ICT in TOT

15.1 Meaning of mass contact methods

Mass contact methods are teaching methods that reach **many people at the same time**, often an entire village, block or district. The contact is largely **impersonal** - the extension worker does not meet each person individually. Mass methods are the fastest and cheapest way to **create awareness** and to spread information widely, but they are weak for teaching skills or for convincing an individual.

15.2 Important mass contact methods

15.2.1 Campaign

A campaign is an **intensive, concentrated effort** to spread a single message among a large number of people within a short time and a definite area, using many methods together. For example, a "seed treatment campaign" or a "soil testing campaign" in a block. A campaign creates wide awareness quickly and builds enthusiasm.

Advantages: Reaches many people quickly; creates awareness and enthusiasm; uses many methods together; good for a single, urgent message. **Limitations:** Short-lived; the message is not deep; follow-up is needed; may create only awareness, not adoption.

15.2.2 Exhibition

An exhibition is a **display** of objects, models, charts, posters and demonstrations arranged so that people can see and learn. Agricultural exhibitions (such as the TNAU and state agricultural exhibitions) show new varieties, implements, products and technologies.

Advantages: Reaches many people; visual and interesting; shows many things at one place; good for creating awareness and interest. **Limitations:** Costly to arrange; the visitor is a spectator, not a learner; follow-up is needed; may be crowded and tiring.

15.2.3 Farmers' day (field day)

A farmers' day, or field day, is a **special day** on which farmers are invited to a farm, a KVK or a demonstration plot to see the results of a technology, to meet experts and to discuss. It is usually held at the time of harvest of a result demonstration.

Advantages: Shows results at first hand; brings farmers and scientists together; creates interest and confidence; good for a specific technology. **Limitations:** Reaches only those who attend; needs a good demonstration to show; one day is short for deep learning.

15.2.4 Field trip

A field trip is a **visit by a group of farmers to a place away from their village** - a research station, a KVK, a progressive farmer's field or a market - to see and learn something new.

Advantages: Broadens the farmer's outlook; shows new practices in real settings; creates interest; farmers learn from other farmers. **Limitations:** Costly and time-consuming; only a few can go; the experience must be followed up.

15.2.5 Other mass methods

- **Radio:** Wide reach, low cost, useful for illiterate listeners; only one-way.
- **Television:** Powerful audio-visual reach; costly to produce; one-way.
- **Newspaper and print media:** Detailed and lasting; only for literates.

- **Leaflets, folders and bulletins:** Cheap printed material; lasting; for literates.
- **Posters and charts:** Quick visual message; good for awareness.
- **Internet and social media:** Fast, wide and two-way; needs a device and connectivity.

15.3 Comparison of mass contact methods

Method	Reach	Cost per person	Main use	Limitation
Campaign	Very wide	Low	Awareness of one message	Short-lived; needs follow-up
Exhibition	Wide	Low-medium	Awareness and interest	Spectator, not learner
Farmers' day	Medium	Medium	Show results	Only attendees benefit
Field trip	Small group	High per person	Learning from other places	Few can go
Radio / TV	Very wide	Very low	Awareness and information	One-way
Print media	Wide among literates	Low	Information and detail	Literates only

15.4 ICT applications in transfer of technology

Information and communication technology (ICT) has become a major tool of transfer of technology. The main applications are:

- **New media:** Internet portals, mobile applications, e-learning modules, video conferencing, podcasts, webinars and online advisory services.
- **Social media:** WhatsApp groups, Facebook pages, YouTube channels, Instagram and Twitter (X) for quick sharing of advisories, photos and short videos.
- **Mobile applications:** Uzhavan app (Tamil Nadu), m-Velanmai, Kisan Suvidha, Pusa Krishi and others.
- **Call centres and helplines:** Kisan Call Centre gives free advice on a toll-free number in the local language.
- **Expert systems and decision-support tools:** For disease diagnosis and fertiliser recommendation.
- **Remote sensing and weather services:** For crop and weather advisories.

15.4.1 Advantages of ICT in TOT

- Fast, wide and low-cost reach.
- Two-way communication - the farmer can ask questions.
- Timely alerts on weather, pest and price.
- Multimedia suits learners of different levels, including low-literacy learners.
- Records of queries and advisories help in planning.

15.4.2 Limitations of ICT in TOT

- Needs device, connectivity and electricity.
- Digital divide - the poor, the elderly and remote areas may be left out.
- Content must be in the local language and locally relevant.
- Trust and credibility must be built.
- Needs trained staff and maintenance.

15.5 Media mix strategy

A **media mix** means the planned **combination of several teaching methods and media** to achieve an objective, because no single method is best for all purposes. Different methods do different jobs, and used together they reinforce one another.

For example, to spread a new paddy variety, the extension worker may use:

- A **campaign** and **posters** to create awareness.
- A **result demonstration** to convince.
- A **method demonstration** and **field day** to teach and show.
- **Farm and home visits** to get adoption from individual farmers.
- A **WhatsApp group** and **radio** for reminders and follow-up.

```
MEDIA MIX - A STAGED APPROACH
Stage of adoption      Methods / media used
---------------------------------------------------------------
Awareness          -> Campaign, posters, radio, TV, social media
Interest           -> Exhibition, farmers' day, leaflet, group meeting
Evaluation         -> Result demonstration, field trip, farm visit
Trial              -> Method demonstration, farm and home visit
Adoption           -> Farm visit, personal letter, group discussion
```

The principle is: **use mass methods for awareness, group methods for teaching, and individual methods for adoption**, and combine them in a planned sequence.

15.6 Worked example - designing a media mix

Problem. A KVK wants to popularise a new short-duration paddy variety in 10 villages within one season. Suggest a media mix and the reason for each choice.

Solution.

Stage	Method / medium	Reason
Awareness	Campaign, posters, radio, WhatsApp	To reach all farmers quickly
Interest	Village meeting, leaflet	To explain the advantages
Evaluation	Result demonstration in 2 villages	To show the yield difference
Trial	Method demonstration, farm visit	To help farmers try the variety
Adoption	Farmers' day, farm and home visits	To confirm and extend adoption

Exam Focus

Key points to remember

- Mass contact methods reach many people at once, impersonally; best for creating awareness, weak for skills.
- Mass methods: campaign, exhibition, farmers' day (field day), field trip, radio, TV, print media.
- Campaign = intensive short-time effort with one message; exhibition = display of objects and models.
- ICT in TOT: new media (apps, webinars, podcasts), social media, call centres, expert systems.
- ICT advantages: fast, wide, cheap, two-way, timely alerts; limitations: connectivity, digital divide, language.
- Media mix = planned combination of methods and media to achieve an objective.
- Rule: mass methods for awareness, group methods for teaching, individual methods for adoption.

Short-answer questions with model answers

Q1. What are mass contact methods? Give two examples.

Ans. Mass contact methods reach many people at the same time through impersonal contact. Examples: campaign, exhibition, farmers' day, field trip, radio and television.

Q2. What is a campaign?

Ans. A campaign is an intensive, concentrated effort to spread a single message among a large number of people within a short time and a definite area, using many methods together.

Q3. What is a media mix?

Ans. A media mix is the planned combination of several teaching methods and media to achieve an objective, because no single method is best for all purposes.

Q4. State two advantages and two limitations of ICT in TOT.

Ans. Advantages: fast, wide, low-cost reach and two-way communication. Limitations: needs device, connectivity and power, and the digital divide may exclude the poor and remote areas.

Long-answer questions with model answer outlines

Q1. Explain the important mass contact methods used in extension education with their merits and limitations.

Outline. Define mass contact methods and state their general merits and limits. Describe campaign, exhibition, farmers' day and field trip in detail, and briefly radio, TV and print media. Present the comparison table. Conclude that mass methods create awareness and must be followed by group and individual methods for adoption.

Q2. Discuss the applications of ICT in transfer of technology and explain the media mix strategy.

Outline. Define ICT and its applications in TOT - new media, social media, mobile apps (Uzhavan, m-Velanmai), call centres, expert systems. Give advantages and limitations. Define media mix and show the staged approach diagram. Solve a media-mix example. Conclude that ICT complements traditional methods and that the media mix must match the stage of adoption.

Chapter 16. Communication and Agricultural Journalism

16.1 Communication: meaning and definition

Communication is the process of passing information, ideas, feelings and attitudes from one person to another, so that the receiver understands the message in the way the sender intended. The word comes from the Latin *communis*, meaning "common" - communication makes something common between two persons.

Definitions:

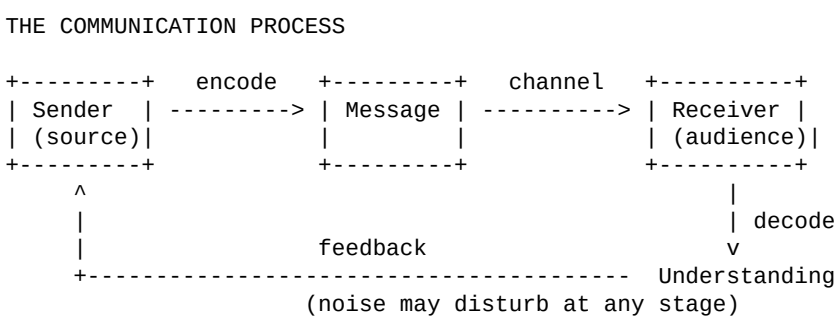
- **Leagans:** Communication is the process by which two or more people exchange ideas, facts, feelings or impressions in ways that each gains a common understanding of the meaning, intent and use of the message.
- **Rogers and Shoemaker:** Communication is the process by which messages are transferred from a source to a receiver.
- **Reddy:** Communication is the process of transmitting information and understanding from one person to another.

The key idea in every definition is **common understanding**. If the message is sent but not understood, there is no communication.

16.2 Elements of the communication process

The communication process has the following elements:

- **Sender (source / communicator):** The person who has the idea and sends the message.
- **Message:** The information, idea or feeling that is sent.
- **Encoding:** Putting the idea into words, symbols or pictures.
- **Channel (medium):** The means by which the message travels - speech, print, radio, mobile.
- **Receiver (audience):** The person who receives the message.
- **Decoding:** The receiver interpreting the message.
- **Feedback:** The receiver's response, which tells the sender whether the message was understood.
- **Noise:** Anything that interferes with the message - physical noise, language, distance, distraction.



16.3 Principles of communication

Good communication follows certain principles:

- **Principle of clarity:** The message must be clear and simple; avoid difficult words.
- **Principle of completeness:** The message must be complete, with all that the receiver needs to act.
- **Principle of conciseness:** Say it briefly; long messages are forgotten.
- **Principle of concreteness:** Use facts, examples and figures, not vague statements.
- **Principle of correctness:** The information must be accurate and up to date.
- **Principle of courtesy:** Be polite and respectful; do not offend the receiver.
- **Principle of the audience:** Use the language, level and interest of the receiver.
- **Principle of two-way communication:** Encourage feedback; listen as well as speak.
- **Principle of appropriate channel:** Choose the medium that the receiver can use.
- **Principle of repetition and reinforcement:** Repeat the message through more than one method.
- **Principle of credibility:** The receiver must trust the sender and the message.
- **Principle of participation:** Involve the receiver in the communication.

16.4 Functions of communication

- **Information function:** To give facts, news and knowledge.
- **Education / instruction function:** To teach skills and change knowledge.
- **Persuasion function:** To convince and to change attitudes and behaviour.
- **Entertainment function:** To interest and hold the attention of the audience.
- **Coordination function:** To bring people together and to coordinate their work.
- **Motivation function:** To encourage people to act.
- **Feedback function:** To bring the receiver's response back to the sender.
- **Social function:** To maintain social relations and community feeling.

16.5 Types of communication

Basis	Types	Meaning
Direction	One-way / two-way	Without or with feedback
Number	Intrapersonal	Within oneself (thinking)
Number	Interpersonal	Between two persons (face-to-face)
Number	Group	Within a group
Number	Mass	To many people through media
Medium	Verbal	Using words - spoken or written
Medium	Non-verbal	Using gestures, expressions, pictures
Formality	Formal	Official, structured (letter, report)
Formality	Informal	Casual, personal (chat, gossip)

16.6 Barriers to communication and how to remove them

Barrier	Example	Remedy
Physical	Noise, distance, poor network	Choose a quiet place; use a suitable medium
Language	Difficult words, dialect difference	Use simple local language
Psychological	Fear, prejudice, distrust	Build trust; be friendly
Cultural	Customs and beliefs	Respect culture; adapt the message
Perceptual	Seeing only what one expects	Use examples and demonstrations
Organisational	Too many levels, delay	Reduce levels; keep messages clear

16.7 Agricultural journalism (print media)

16.7.1 Definition

Agricultural journalism is the branch of journalism that deals with the collection, writing, editing and dissemination of **agricultural information and news** to farmers and others through the print media (newspapers, magazines, bulletins, leaflets) and, in modern usage, also through electronic media. It is the use of the journalist's craft to serve agriculture.

In simple words, agricultural journalism is the **reporting and writing of agricultural news and information** for the benefit of the farming community.

16.7.2 Principles of agricultural journalism

- **Accuracy:** Facts must be correct; a wrong figure or advice can harm crops and incomes.
- **Clarity:** Write in simple language that the farmer can understand.
- **Conciseness:** Be brief; the farmer has little time to read.
- **Objectivity and fairness:** Present the facts without bias; do not favour a product or a company.
- **Relevance and timeliness:** Write what is useful to the farmer now, in the right season.
- **Interest:** Use a good "lead" and human interest to attract the reader.
- **Utility and service:** The purpose is to help the farmer; every item must be useful.
- **Verification:** Cross-check technical facts with scientists before publishing.
- **Localisation:** Use local names, examples and conditions.

16.7.3 Importance of agricultural journalism

- It carries scientific information to a large number of farmers at low cost.
- It gives timely information on weather, pests, prices and government schemes.
- It provides a permanent record that the farmer can read again and again.
- It links farmers, scientists, extension workers, policy-makers and the market.
- It creates awareness and interest, which supports the work of extension.
- It gives farmers a voice through letters, queries and success stories.
- It supports agricultural marketing and rural development.

16.7.4 ABC of news (news writing)

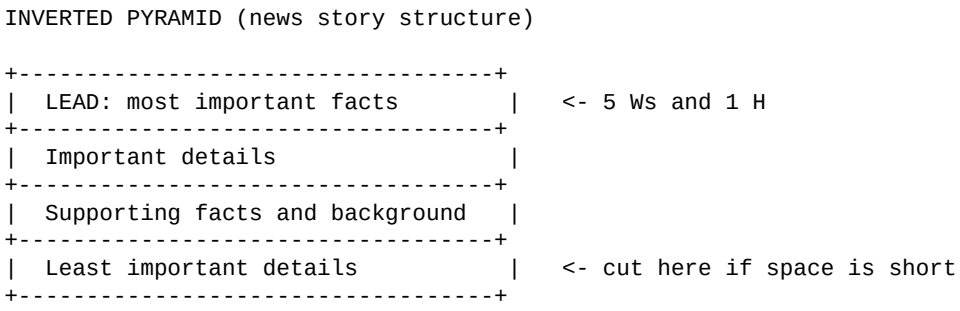
The "ABC of news" refers to the three basic qualities that news writing must have. They are often remembered as **A-B-C**:

- **A - Accuracy:** Every fact, name, figure and date must be correct. Accuracy is the first law of journalism.
- **B - Brevity:** News must be short and to the point; unnecessary words must be cut. "Brevity is the soul of wit."
- **C - Clarity:** The news must be clear and easy to understand, in simple language, so that the reader gets the message at once.

Some teachers also add **D - Depth** (adequate background) and **E - Empathy** (understanding the reader), but the standard ABC of news is **Accuracy, Brevity and Clarity**.

16.7.5 Structure of a news story - the "lead" and the inverted pyramid

A news story begins with a **lead** (or intro) - the first sentence or paragraph that gives the most important facts: **what, who, when, where, why and how** (the "5 Ws and 1 H"). The details follow in decreasing order of importance, so that the story can be cut from the bottom if space is short. This structure is called the **inverted pyramid**.



16.7.6 Types of news

Type of news	Meaning / example
Straight news	Factual report of an event - a farmers' day, a crop loss
Interpretative news	Facts explained with background and meaning - why prices rose
Feature	A longer, human-interest article - a success story of a farmer
News story / event	Reporting a happening - an agricultural exhibition
Advisory / service news	Timely advice - "spray now to control this pest"
Success story	The achievement of a farmer, a KVK or an SHG
Investigative	Deep study of a problem - misuse of a subsidy
Weather and market news	Daily price and weather information
Editorial	The paper's own opinion on an agricultural issue
Scheme news	Information about a government programme

16.7.7 Print media used in agricultural extension

- **Newspapers and farm pages:** Daily papers carry farm news and advisories.
- **Farm magazines:** Weekly or monthly magazines such as *Indian Farming*, *Kisan World*, *The Hindu - Survey of Indian Agriculture*.

- **Leaflets and folders:** Single-sheet, one-topic printed material.
- **Bulletins and pamphlets:** Longer printed material on a subject.
- **Posters and charts:** Visual printed material for awareness.
- **Extension newsletters:** Publications of KVKs, ATMA and the Department of Agriculture.

Exam Focus

Key points to remember

- Communication = process of exchanging ideas and information so that each gains common understanding.
- Elements: sender, message, encoding, channel, receiver, decoding, feedback, noise.
- Principles: clarity, completeness, conciseness, concreteness, correctness, courtesy, audience, two-way, channel, repetition, credibility, participation.
- Functions: information, education, persuasion, entertainment, coordination, motivation, feedback, social.
- Barriers: physical, language, psychological, cultural, perceptual, organisational - and their remedies.
- Agricultural journalism = collection, writing and dissemination of agricultural news through print media.
- ABC of news = Accuracy, Brevity, Clarity.
- News structure: the lead (5 Ws and 1 H) and the inverted pyramid; types of news (straight, interpretative, feature, advisory, success story, etc.).

Short-answer questions with model answers

Q1. Define communication.

Ans. Communication is the process by which two or more people exchange ideas, facts, feelings or impressions so that each gains a common understanding of the meaning, intent and use of the message (Leagans).

Q2. Name the elements of the communication process.

Ans. Sender, message, encoding, channel, receiver, decoding, feedback and noise.

Q3. Define agricultural journalism.

Ans. Agricultural journalism is the collection, writing, editing and dissemination of agricultural information and news to farmers and others through the print media (and modern electronic media).

Q4. What is the ABC of news?

Ans. The three basic qualities of news writing - Accuracy (correct facts), Brevity (short and to the point) and Clarity (simple and easy to understand).

Long-answer questions with model answer outlines

Q1. Explain the meaning, elements, principles and functions of communication, and discuss the barriers to communication.

Outline. Define communication with two definitions. Explain the elements with the process diagram. List and explain the principles and the functions. Describe the barriers with the table and their remedies. Conclude that effective communication requires common understanding and feedback.

Q2. What is agricultural journalism? Explain its principles, importance, the ABC of news and the types of news.

Outline. Define agricultural journalism. Explain its principles (accuracy, clarity, brevity, objectivity, relevance, etc.) and importance. Explain the ABC of news - accuracy, brevity, clarity. Describe the lead and the inverted pyramid with the diagram. List the types of news with the table. Conclude with the role of print media in extension.

Chapter 17. Diffusion and Adoption of Innovation

17.1 Concept and meaning of diffusion

Diffusion is the process by which a new idea or practice is communicated through certain channels over time among the members of a social system. In simple terms, diffusion is the **spread of a new practice** from person to person, from the first few who adopt it to the many who follow.

Adoption is the decision by an individual to make full use of a new idea or practice as the best course of action available. Diffusion is the **spread** of the practice in the community; adoption is the **decision** of an individual to use it.

17.2 Definitions of diffusion and adoption

- **Everett M. Rogers:** Diffusion is the process by which an innovation is communicated through certain channels over time among the members of a social system.
- **Rogers:** Adoption is a decision to make full use of an innovation as the best course of action available.
- **Lionberger:** Adoption is the acceptance and continued use of a new idea or practice by an individual.
- **Diffusion (in extension):** Diffusion is the social process by which an innovation spreads among the people of a community over time.

The four elements of diffusion are: the **innovation** (new idea), the **communication channels**, **time** and the **social system**.

17.3 Attributes of an innovation

The rate at which an innovation is adopted depends largely on five attributes (characteristics) of the innovation itself.

- **Relative advantage:** The degree to which the innovation is perceived as better than the practice it replaces - in profit, yield, saving of labour or prestige. The greater the relative advantage, the faster the adoption.
- **Compatibility:** The degree to which the innovation is consistent with the existing values, beliefs, customs and needs of the people. An innovation that fits the culture is adopted faster.
- **Complexity:** The degree to which the innovation is difficult to understand and use. Simple innovations are adopted faster; complex ones slowly.
- **Trialability:** The degree to which the innovation can be tried on a small scale. Innovations that can be tried in a small plot are adopted faster.
- **Observability:** The degree to which the results of the innovation are visible to others. Innovations whose results can be seen are adopted faster.

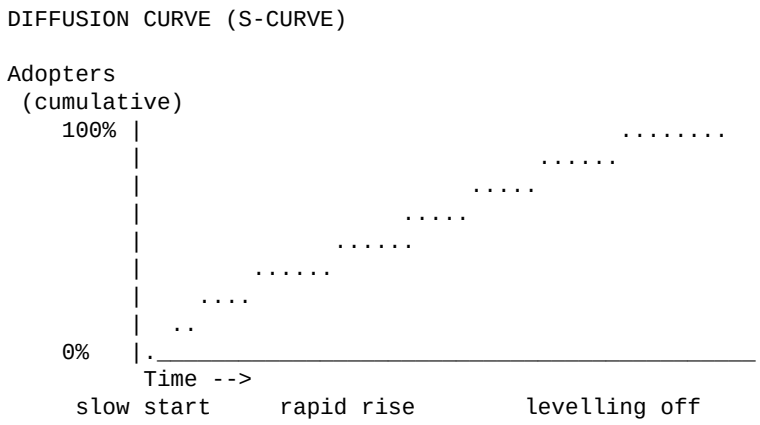
Attribute	Meaning	Effect on adoption
Relative advantage	Better than the old practice	Higher advantage -> faster
Compatibility	Fits values and needs	More compatible -> faster
Complexity	Difficult to understand/use	More complex -> slower
Trialability	Can be tried on a small scale	Easier to try -> faster

Attribute	Meaning	Effect on adoption
Observability	Results are visible	More visible -> faster

Memory aid: **R-C-C-T-O** - Relative advantage, Compatibility, Complexity, Trialability, Observability. Of these, relative advantage and compatibility most influence the rate of adoption.

17.4 Diffusion curve (S-curve)

When the adoption of an innovation is plotted against time, the curve is slow at first (few adopters), rises rapidly in the middle (most adopters), and then levels off (few remaining adopters). This shape is called the **S-curve** or the diffusion curve.



17.5 The innovation-decision process (stages of adoption)

Adoption is not a single act; it is a process that passes through several stages. The five widely used stages are:

1. **Awareness (Knowledge):** The farmer first learns that the new practice exists, but knows little about it. The extension worker's job here is to create awareness.
2. **Interest (Persuasion):** The farmer becomes interested and wants more information. He seeks details and forms a favourable or unfavourable attitude.
3. **Evaluation:** The farmer mentally weighs the advantages and disadvantages, and decides whether to try it. "Will it work for me?"
4. **Trial:** The farmer tries the practice on a small scale to see how it works. This is the most important stage for the extension worker, who should give help and materials.
5. **Adoption:** The farmer decides to use the practice fully and continues to use it. If he later gives it up, it is **discontinuance**.

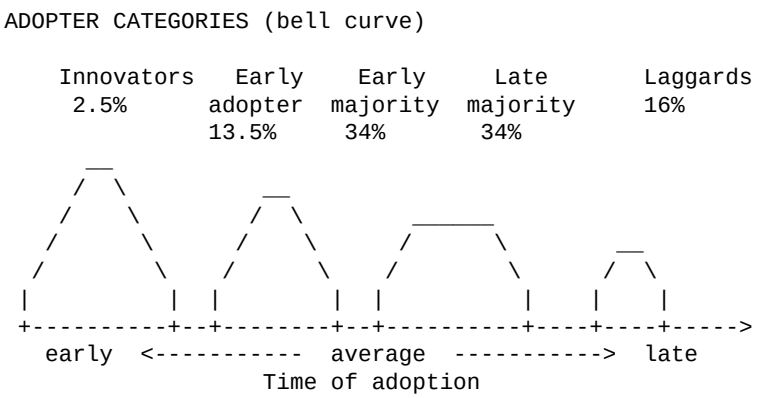
Stage	What the farmer does	Extension worker's task
Awareness	Learns that the practice exists	Create awareness - mass methods
Interest	Wants more information	Give information - group methods
Evaluation	Weighs pros and cons	Convince - result demonstration
Trial	Tries on a small scale	Help with inputs and advice - method demonstration, visit
Adoption	Uses it fully and continues	Confirm, follow up, extend to others

Note: Some texts describe the stages as Awareness, Interest, Evaluation, Trial and Adoption (the "A-I-E-T-A" five stages). A more recent version adds a final stage of "Confirmation".

17.6 Adopter categories

People in a social system adopt an innovation at different times. Based on how early they adopt, farmers are grouped into five **adopter categories**. Rogers used the normal distribution (bell curve) to define them.

- **Innovators (about 2.5%):** Venturesome; the first to adopt; willing to take risks; often young and better educated; may be regarded as "gamblers".
- **Early adopters (about 13.5%):** Respectable; the leaders of the community; adopt early and are looked up to by others; the key group for the extension worker to work with.
- **Early majority (about 34%):** Deliberate; adopt just before the average farmer; not leaders but influential; they follow the early adopters.
- **Late majority (about 34%):** Skeptical; adopt after the average; adopt mainly because of social pressure and economic necessity; the largest group with the early majority.
- **Laggards (about 16%):** Traditional; the last to adopt; suspicious of change; often older, with small resources; they adopt only when forced by necessity.



Category	% of adopters	Key traits	Extension approach
Innovators	2.5	Venturesome, risk-taking, young	Give new technology first
Early adopters	13.5	Respected leaders, opinion leaders	Work with them to demonstrate
Early majority	34	Deliberate, follow leaders	Teach and encourage
Late majority	34	Skeptical, follow pressure	Provide support and remove doubts
Laggards	16	Traditional, suspicious, old	Patient, repeated contact

17.7 Worked example - adoption percentage and diffusion rate

Problem. In a village of 200 farmers, a new wheat variety was introduced. By the end of the first year, 10 farmers had adopted it; by the end of the second year, 60 farmers; by the end of the third year, 140 farmers; and by the end of the fourth year, 180 farmers. Calculate the adoption percentage at the end of each year and identify the stage of the diffusion curve.

Solution.

Year	Adopters	Adoption % = (adopters / 200) x 100	Stage of the S-curve
1	10	5.0%	Slow start (innovators)
2	60	30.0%	Rising
3	140	70.0%	Rapid rise (majority adopt)
4	180	90.0%	Levelling off (laggards)

Interpretation: The adoption percentage rose from 5% to 90% in four years, giving the typical S-shape - a slow start, a rapid rise in years 2-3, and a levelling off in year 4. The rapid rise in the middle shows the influence of the early adopters and the early majority.

17.8 Factors affecting the rate of diffusion and adoption

- **Attributes of the innovation:** Relative advantage, compatibility, complexity, trialability, observability.
- **Characteristics of the adopter:** Age, education, farm size, income, risk-bearing ability, social participation.
- **Communication:** The channels used - mass media for awareness, personal and local channels for adoption.
- **Social system:** The norms, values, leadership and social structure of the community.
- **Extension effort:** The number and quality of demonstrations, visits and contacts.
- **Economic factors:** Cost of the practice, availability of credit and market.
- **Physical factors:** Irrigation, soil, climate and infrastructure.

17.9 Consequences and discontinuance

Adoption may not always continue. **Discontinuance** is the decision to stop using a practice after having adopted it. It may be due to:

- **Dissatisfaction discontinuance:** The practice did not give the expected result.
- **Replacement discontinuance:** The farmer replaced it with a better practice.

Exam Focus

Key points to remember

- Diffusion = spread of a new idea through channels over time in a social system; adoption = an individual's decision to use it.
- Four elements of diffusion: innovation, communication channels, time, social system.
- Attributes of innovation: relative advantage, compatibility, complexity, trialability, observability (R-C-C-T-O).
- Diffusion follows an S-curve - slow start, rapid rise, levelling off.
- Stages of adoption: awareness, interest, evaluation, trial, adoption (A-I-E-T-A); some add confirmation.
- Adopter categories: innovators (2.5%), early adopters (13.5%), early majority (34%), late majority (34%), laggards (16%).
- Early adopters are the opinion leaders and the key group for the extension worker.

- Discontinuance = stopping the use of a practice after adoption.

Short-answer questions with model answers

Q1. Define diffusion and adoption.

Ans. Diffusion is the process by which an innovation is communicated through certain channels over time among the members of a social system (Rogers). Adoption is the decision to make full use of an innovation as the best course of action available.

Q2. Name the five attributes of an innovation.

Ans. Relative advantage, compatibility, complexity, trialability and observability.

Q3. What are the stages of adoption?

Ans. Awareness, interest, evaluation, trial and adoption (some texts add confirmation).

Q4. Name the adopter categories with their percentages.

Ans. Innovators (2.5%), early adopters (13.5%), early majority (34%), late majority (34%) and laggards (16%).

Long-answer questions with model answer outlines

Q1. Explain the concept of diffusion and adoption, the attributes of an innovation and the stages of the adoption process.

Outline. Define diffusion and adoption; state the four elements. Explain the five attributes with the table and their effect on adoption. Describe the five stages of adoption with the extension worker's task at each stage. Draw the S-curve and explain its three phases. Conclude that understanding these helps the extension worker to speed up adoption.

Q2. Describe the adopter categories and the diffusion curve, and explain the factors affecting the rate of adoption.

Outline. Define diffusion and draw the S-curve. Explain the five adopter categories with percentages and traits (bell-curve diagram and table). Explain the factors affecting the rate of adoption - attributes of the innovation, adopter characteristics, communication, social system, extension effort, economic and physical factors. Solve the adoption-percentage example. Conclude with the role of early adopters as opinion leaders.

Glossary

- **Adoption:** The decision by an individual to make full use of a new idea or practice as the best course of action available, and to continue using it.
- **Adopter categories:** The five groups of farmers - innovators, early adopters, early majority, late majority and laggards - classified by how early they adopt an innovation.
- **Agricultural journalism:** The collection, writing, editing and dissemination of agricultural information and news to farmers through the print (and electronic) media.
- **ATMA:** Agricultural Technology Management Agency - a district-level, decentralised body for coordinating and managing agricultural extension.
- **Campaign:** An intensive, concentrated effort to spread a single message among many people in a short time and a definite area.
- **Capacity building:** The process of developing the knowledge, skills, attitudes and abilities of extension personnel, mainly through training.
- **Change agent:** The professional extension worker whose job is to bring about desirable change, usually by working through local leaders.
- **Communication:** The process of exchanging ideas, facts, feelings or impressions so that each person gains a common understanding of the message.
- **Community development:** The process by which the people's own efforts are united with those of government or other agencies to improve the economic, social and cultural conditions of the whole community.
- **Compatibility:** The degree to which an innovation is consistent with the existing values, beliefs, customs and needs of the people.
- **Complexity:** The degree to which an innovation is difficult to understand and use.
- **Contact farmer:** A progressive farmer selected in the T&V system as the first point of contact for extension advice.
- **Cyber extension (e-extension):** The use of information and communication technology to deliver extension information and advisory services to farmers.
- **Demonstration:** A teaching method that shows a practice or its results; a method demonstration shows "how", a result demonstration shows "what happens".
- **Diffusion:** The process by which an innovation is communicated through certain channels over time among the members of a social system.
- **Discontinuance:** The decision to stop using a practice after having adopted it.
- **DWCRA:** Development of Women and Children in Rural Areas - a programme organising rural women into groups for income generation.
- **Early adopter:** The adopter category (about 13.5%) of respected, influential farmers who adopt early and act as opinion leaders.
- **Education:** The deliberate and systematic process of bringing about desirable changes in the knowledge, skill and attitude of a person.
- **Expert system:** A computer program that imitates a human expert by using a knowledge base and an inference engine to give reasoned advice.

- **Extension education:** Out-of-school, non-formal education for rural people, carried to their doorstep, that helps them to help themselves.
- **e-Velanmai:** An ICT-based, participatory extension advisory system of Tamil Nadu in which a trained farmer sends fellow farmers' problems to a panel of scientists.
- **Farmer Field School (FFS):** A participatory group-learning method in which farmers learn by observing and experimenting in their own fields and teach one another.
- **Farmer-led extension:** A participatory approach in which farmers themselves take the lead in generating and sharing agricultural knowledge.
- **Feedback:** The receiver's response in communication, which tells the sender whether the message was understood.
- **Firka Development Scheme:** A pre-independence integrated rural development scheme of the Madras Government (1947-48) using the firka as the unit.
- **Formal education:** Institutionalised education with a fixed syllabus, timetable, examinations and certification.
- **Informal education:** Natural, unplanned, lifelong learning from family, society and experience, without certification.
- **Innovation:** A new idea, practice or object perceived as new by an individual or a social system.
- **Innovators:** The adopter category (about 2.5%) of venturesome, risk-taking farmers who adopt first.
- **KVK:** Krishi Vigyan Kendra (Farm Science Centre) - a district-level institution for vocational training, demonstrations and advisory services.
- **Laggards:** The adopter category (about 16%) of traditional, suspicious farmers who adopt last.
- **Leadership:** The process of influencing people to cooperate willingly towards a group goal.
- **Market-led extension:** A demand-driven approach that helps farmers produce according to market demand and market their produce profitably.
- **Mass contact methods:** Teaching methods that reach many people at once through impersonal contact, such as campaign and exhibition.
- **Media mix:** The planned combination of several teaching methods and media to achieve an objective.
- **Method demonstration:** A group teaching method that shows "how to do" a practice step by step (learning by doing).
- **Monitoring:** The continuous and systematic checking of the progress of an ongoing programme to take timely corrective action.
- **Non-formal education:** Organised but flexible, voluntary and life-centred education outside the formal system; extension education is non-formal.
- **Observability:** The degree to which the results of an innovation are visible to others.
- **Opinion leader:** A person whom others consult for advice on a topic, who is among the first to be influenced and the first to influence others.
- **PMGSY:** Pradhan Mantri Gram Sadak Yojana - a programme providing all-weather road connectivity to unconnected villages.

- **Privatisation of extension:** The provision of extension services by private agencies, often for a fee or linked to the sale of a product.
- **Programme planning:** The process of determining objectives, establishing a course of action, deciding methods and evaluating results, with the participation of the people.
- **Relative advantage:** The degree to which an innovation is perceived as better than the practice it replaces.
- **Result demonstration:** A demonstration that shows the results of a practice by comparing an improved plot with a control plot (learning by seeing).
- **Rural development:** A strategy and process to improve the economic and social life of the rural poor, covering agriculture, infrastructure, institutions, health and employment.
- **Rural leadership:** The process of influencing rural people voluntarily to bring about desirable changes in their farm, home and community life.
- **Self-Help Group (SHG):** A small group of poor people, usually women, who save together and lend to their members, used as the base of many development programmes.
- **T&V system:** Training and Visit system (Reorganised Extension System) - a system of regular training of extension workers and fixed visits to contact farmers.
- **Teaching method:** The device by which the extension worker brings the farmer and the recommended practice together so that the farmer learns and adopts it.
- **TOT (Transfer of technology):** The process by which research results are carried to the farmer and adopted in the field; a continuous two-way process.
- **Training:** A planned and systematic process of developing the knowledge, skills and attitudes of a person for a particular job.
- **Trialability:** The degree to which an innovation can be tried on a small scale.
- **Uzhavan app:** A Tamil Nadu mobile application giving farmers crop advisories, weather, market prices, machinery booking and subsidy information.
- **VLW:** Village Level Worker - the multi-purpose extension worker at the village level under the Community Development Programme.

References

The following standard texts and sources form the basis of the material in this handbook. Students are advised to consult them for deeper reading and for the exact wording of definitions.

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2 Van den Ban, A. W. and Hawkins, H. S. *Agricultural Extension.* Longman Scientific and Technical, Harlow, UK.

3 Ahire, R. D. *Extension Education and Rural Development.* B. R. Publishing Corporation, Delhi.

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5 Dahama, O. P. and Bhatnagar, O. P. *Education and Communication for Development.* Oxford and IBH Publishing Co., New Delhi.

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9 Leagans, J. P. *Extension Education for Community Development.* (Source of the standard definition of extension education.)

10 Rogers, E. M. *Diffusion of Innovations.* The Free Press, New York. (Source of the diffusion, adopter category and innovation-attribute concepts.)

11 Singh, R. *Rural Development - Principles, Policies and Management.* Sage Publications, New Delhi.

12 ICAR e-course material on "Fundamentals of Extension Education" and related courses, Indian Council of Agricultural Research, New Delhi.

13 TNAU e-course material and lecture notes on AEX 201 - Fundamentals of Extension Education, Tamil Nadu Agricultural University, Coimbatore.

14 Department of Agriculture, Government of Tamil Nadu - scheme documents and the Uzhavan mobile application portal.

15 Ministry of Rural Development, Government of India - programme guidelines (MGNREGA, PMGSY, NRLM and related schemes).

16 Ministry of Agriculture and Farmers' Welfare, Government of India - agricultural extension reform and ATMA documents.

A note on sources: This handbook is a study aid prepared from the standard extension-education literature listed above. Where a programme's exact year or outlay was not certain, the programme has been described in general terms. For any figure quoted in an examination, always verify it against the latest official document.

Appendix A. Full Forms of Abbreviations

This appendix lists the full forms of every abbreviation used in this handbook. Learning the full forms is the fastest way to gain marks in the examination.

Abbreviation	Full form
AEX	Agricultural Extension (course code)
ATMA	Agricultural Technology Management Agency
BDO	Block Development Officer
BPL	Below Poverty Line
CDP	Community Development Programme
DWCRA	Development of Women and Children in Rural Areas
FFS	Farmer Field School
FLD	Frontline Demonstration
FPO	Farmer Producer Organisation
HYV / HYVP	High Yielding Variety / High Yielding Varieties Programme
IADP	Intensive Agricultural District Programme
IAAP	Intensive Agricultural Area Programme
IAY	Indira Awaas Yojana
ICAR	Indian Council of Agricultural Research
ICT	Information and Communication Technology
IRDP	Integrated Rural Development Programme
ITDA	Integrated Tribal Development Agency
ITDP	Integrated Tribal Development Project
IVLP	Institute Village Linkage Programme
KVK	Krishi Vigyan Kendra
MGNREGA	Mahatma Gandhi National Rural Employment Guarantee Act
MSY	Mahila Samridhi Yojana
NAIP	National Agricultural Innovation Project
NATP	National Agricultural Technology Project
ND	National Demonstration
NES	National Extension Service
NGO	Non-Governmental Organisation
NRLM	National Rural Livelihoods Mission (Aajeevika)
ORP	Operational Research Project
PMGSY	Pradhan Mantri Gram Sadak Yojana
POSDCORB	Planning, Organising, Staffing, Directing, Coordinating, Reporting, Budgeting
REP	Research-Extension-Producer
RMK	Rashtriya Mahila Kosh

Abbreviation	Full form
RSVY	Rashtriya Sam Vikas Yojana
SGSY	Swarnjayanti Gram Swarozgar Yojana
SGRY	Sampoorna Grameen Rozgar Yojana
SHG	Self-Help Group
T&V	Training and Visit (system)
TNAU	Tamil Nadu Agricultural University
TOT	Transfer of Technology
VLW	Village Level Worker
VEW	Village Extension Worker

Appendix B. Quick Revision - All Comparison Tables at a Glance

This appendix collects the most important comparisons of the syllabus in one place. Revise these tables the night before the examination.

B.1 Types of education

Type	Where	Certification	Nature
Formal	School, college	Yes	Rigid, fixed syllabus
Informal	Family, society	No	Unplanned, lifelong
Non-formal	Camps, KVK, extension	Usually no	Organised but flexible

B.2 Pre- and post-independence extension efforts

Era	Effort	Founder	Key idea
Pre	Sriniketan	Rabindranath Tagore	All-round village reconstruction
Pre	Marthandam	Spencer Hatch	Self-help with a friendly hand
Pre	Firka Development Scheme	Madras Government	Firka as unit of development
Pre	Gurgaon Experiment	F. L. Brayne	Official rural uplift; people's will needed
Post	Etawah Pilot Project	Albert Mayer	Extension approach with the people
Post	Nilokheri Experiment	S. K. Dey	Self-sufficient township
Post	T&V system	World Bank / Benor	Regular training and fixed visits

B.3 Individual, group and mass contact methods

Class	Reach	Contact	Best for
Individual	One person	Personal	Convincing and adoption
Group	A group	Face-to-face group	Teaching skills and discussion
Mass	Many people	Impersonal	Awareness and information

B.4 Models of transfer of technology

Model	Flow	Farmer's role
Top-down	Research to extension to farmer	Passive receiver
Feedback	Top-down with feedback loop	Receiver who gives feedback
Farmer-back-to-farmer	Farmer to researcher to farmer	Active partner
REP interaction	Among research, extension and producer	Equal partner

B.5 Types of evaluation

Basis	Type
Time	Ex-ante, concurrent, ex-post
Purpose	Formative, summative
Evaluator	Internal, external
Scope	Process, impact
Comparison	Before-and-after, with-and-without

B.6 Attributes of innovation

Attribute	Meaning	Effect on adoption
Relative advantage	Better than the old practice	Higher -> faster
Compatibility	Fits values and needs	More -> faster
Complexity	Difficult to use	More -> slower
Trialability	Can be tried on a small scale	Easier -> faster
Observability	Results are visible	More -> faster

B.7 Adopter categories

Category	%	Trait
Innovators	2.5	Venturesome, risk-taking
Early adopters	13.5	Respected opinion leaders
Early majority	34	Deliberate, follow leaders
Late majority	34	Skeptical, follow pressure
Laggards	16	Traditional, suspicious

Appendix C. Model Question Papers

C.1 Model Question Paper 1 (full syllabus)

Time: Three hours. Maximum marks: 100.

Part A - Answer any ten of the following (2 marks each, 20 marks).

1 Define extension education and give one standard definition.

2 Differentiate formal and non-formal education.

3 What is a result demonstration?

4 Expand IADP, HYVP and KVK.

5 What is market-led extension?

6 Define rural leadership.

7 What is an S-curve?

8 Name the five attributes of an innovation.

9 Differentiate monitoring and evaluation.

10 What is the ABC of news?

11 Name any four adopter categories with their percentages.

12 What is a media mix?

Part B - Answer any five of the following (6 marks each, 30 marks).

1 Explain the objectives and principles of extension education.

2 Describe the steps in extension programme planning with a diagram.

3 Explain the Reorganised Extension System (T&V) with its advantages and limitations.

4 Explain the meaning and importance of privatisation of extension and cyber extension.

5 Describe the types of leaders in the rural context and the methods of identifying a rural leader.

6 Explain the stages of the adoption process and the extension worker's task at each stage.

Part C - Answer any two of the following (10 marks each, 20 marks).

1 Explain the meaning, scope, process and principles of extension education, and compare it with formal education.

2 Describe the important extension efforts in the pre-independence and post-independence eras in India.

3 Explain the classification of extension teaching methods and describe any five group contact methods.

Part D - Answer any one of the following (30 marks).

1 Explain the concept of diffusion and adoption, the attributes of an innovation, the stages of adoption and the adopter categories, with diagrams.

2 Explain the meaning, principles and functions of communication and discuss the principles, importance and types of news in agricultural journalism.

C.2 Model Question Paper 2 (mid semester)

Time: One and a half hours. Maximum marks: 50.

Part A - Answer any five (2 marks each, 10 marks).

1 What are the three types of education?

2 Define programme planning.

3 What was the motto of the Marthandam project?

4 Expand SGSY and NRLM.

5 What is an expert system?

6 Define rural development.

7 Name any two rural development programmes.

Part B - Answer any three (5 marks each, 15 marks).

1 Explain the fundamental and general objectives of extension education.

2 Describe the eight steps in programme planning with a diagram.

3 Explain the advantages and limitations of the T&V system.

4 Differentiate market-led extension from farmer-led extension.

Part C - Answer any one (10 marks each, 10 marks).

1 Explain the meaning, principles and steps of extension programme planning.

2 Describe the important agriculture development programmes launched by ICAR and the Government of India.

Part D - Answer any one (15 marks).

1 Explain the meaning, concept, principles and philosophy of community development.

2 Describe the important pre-independence and post-independence extension efforts in India.

Answering tips: Write the definition first, then the points with sub-headings, then a table or diagram wherever possible, then an Indian or Tamil Nadu example, and finish with a one-line conclusion. Underline key terms and draw diagrams neatly with a pencil and scale.

Appendix D. Examination Strategy and Answer-Writing Guide

D.1 How to read a question

- Identify the **command word** - define, list, explain, describe, differentiate, justify, or illustrate. Each demands a different kind of answer.
- Note the **marks** - a two-mark answer needs two or three lines, a ten-mark answer needs a definition, points and an example.
- Note the **number of parts** - if the question asks for "meaning, principles and steps", answer all three.

D.2 Structure of a good answer

1. **Definition** - begin with the standard definition of the main term.
2. **Points** - give the points under clear sub-headings or numbering.
3. **Table or diagram** - present a comparison as a table and any process as a diagram; examiners award marks for neat presentation.
4. **Example** - add one Indian or Tamil Nadu example to show application.
5. **Conclusion** - end with a one-line conclusion that answers the question.

D.3 Common mistakes to avoid

- Writing long paragraphs without headings; the examiner cannot find the points.
- Confusing similar terms - monitoring with evaluation, method demonstration with result demonstration, market-led with farmer-led extension.
- Giving wrong years or figures; if unsure, describe the programme in general terms.
- Forgetting the full form of an acronym.
- Omitting the diagram when the question asks for a process.
- Writing everything about the topic without answering the exact question asked.

D.4 High-yield topics for quick revision

- Definitions of education, extension education, community development, communication, diffusion and adoption.
- The comparison tables - types of education, pre- and post-independence efforts, TOT models, types of evaluation, adopter categories.
- The diagrams - extension process, programme planning cycle, communication process, TOT models, S-curve, adopter bell curve.
- The acronyms and their full forms.
- The worked examples - demonstration cost-benefit and adoption percentage.

Final word: Extension education is a subject of ideas and definitions, but it is also a practical subject. Answers that connect theory to a real village, a real KVK or a real Tamil Nadu scheme always score better than answers that only repeat the textbook.